

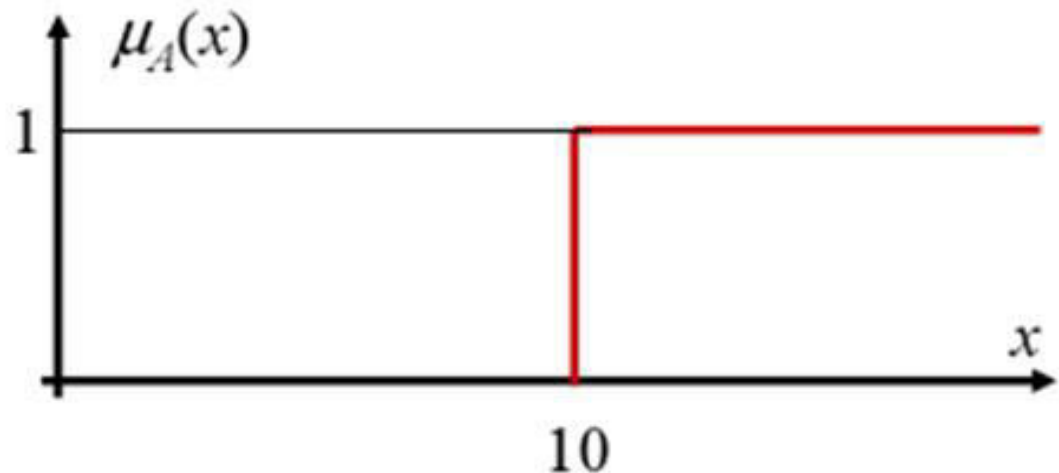
CHAPTER 9

MEMBERSHIP FUNCTIONS

CRISP MEMBERSHIP FUNCTIONS

- Crisp membership functions (μ) are either one or zero.
- Consider the example: Numbers greater than 10. The membership curve for the set A is given by

$$A = \{x \mid x > 10\}$$



MEMBERSHIP FUNCTION

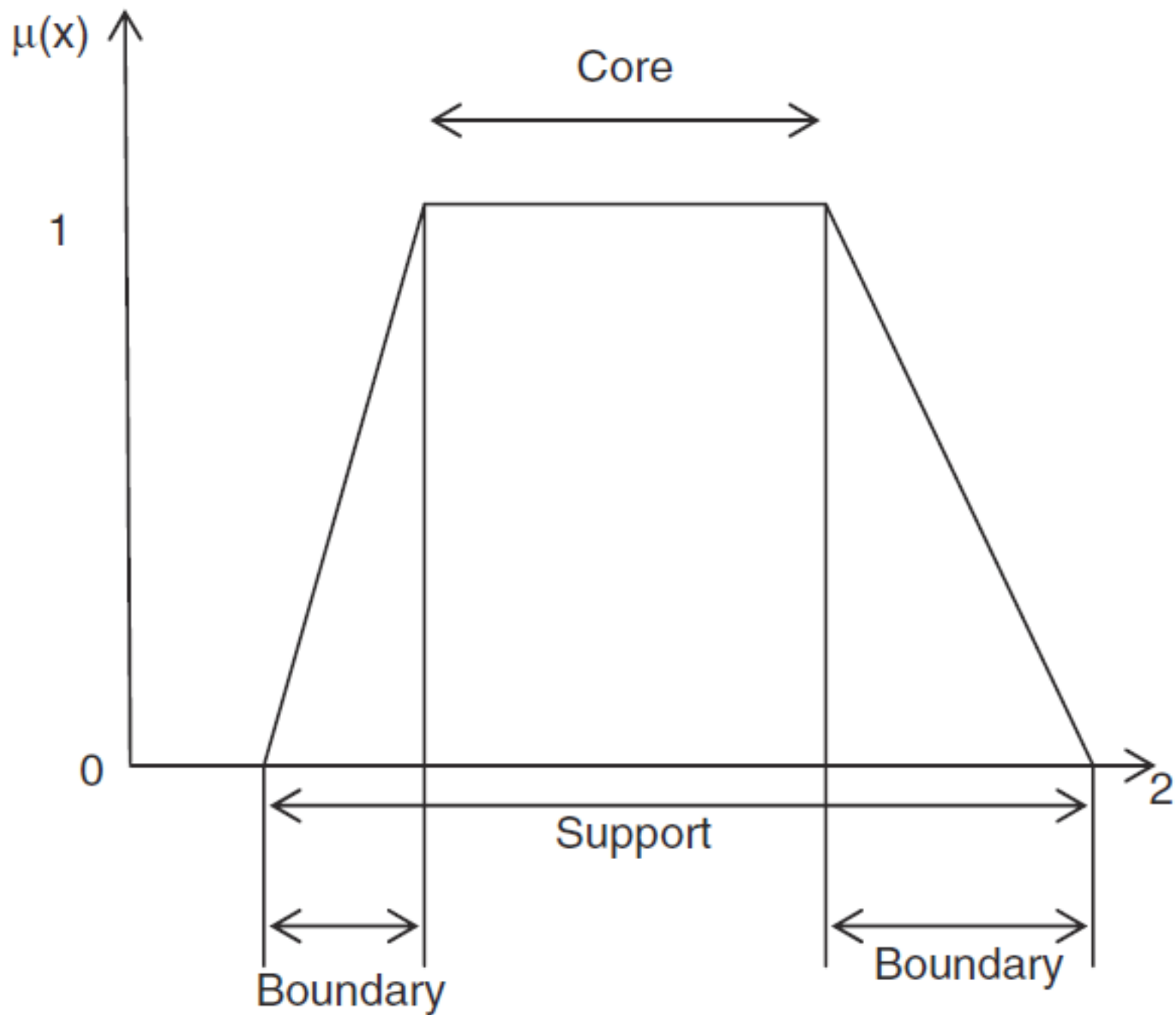
- Membership function defines the fuzziness in the fuzzy set irrespective of the elements in the set, which are discrete or continuous
- Membership functions are generally represented in the graphical form
- There exist certain limitations for the shape used to represent graphical form of membership function
- The rules that describe fuzziness graphically are also fuzzy
- Membership function can be thought of as a technique to solve empirical problems on the basis of experience rather than knowledge

FEATURES OF MEMBERSHIP FUNCTION

- Membership function defines all the information contained in a fuzzy set
- A fuzzy set $A_{\sim}$ in the universe of discourse X can be defined as a set of ordered pairs:

$$A_{\sim} = \{(x, \mu_{A_{\sim}}(x)) \mid x \in X\}$$

- Where $\mu_{A_{\sim}}(x)$ is called membership function of $A_{\sim}$
- The membership function $\mu_{A_{\sim}}(x)$ maps X to the membership space M
- The membership value ranges in the interval $[0,1]$



FEATURES OF MEMBERSHIP FUNCTION

Three main basic features involved in characterizing membership function are following

Core :

The core of the membership function for some fuzzy set $A_{\sim}$ is defined as the region of universe that is characterized by complete membership in the set $A_{\sim}$. The core has elements x of the universe such that

$$\mu_{A_{\sim}}(x) = 1$$

The core of the fuzzy set may be an empty set

Support :

The support is defined as that region of universe that is characterized by a non zero membership in the set $A_{\sim}$. The support comprises the elements x of the universe such that

$$\mu_{A_{\sim}}(x) > 0$$

A fuzzy set whose support is a single element in X with $\mu_{A_{\sim}}(x) = 1$ is referred to as a **fuzzy singleton**

Boundary :

The boundary is defined as that region of universe that is characterized by a non zero but not complete membership.

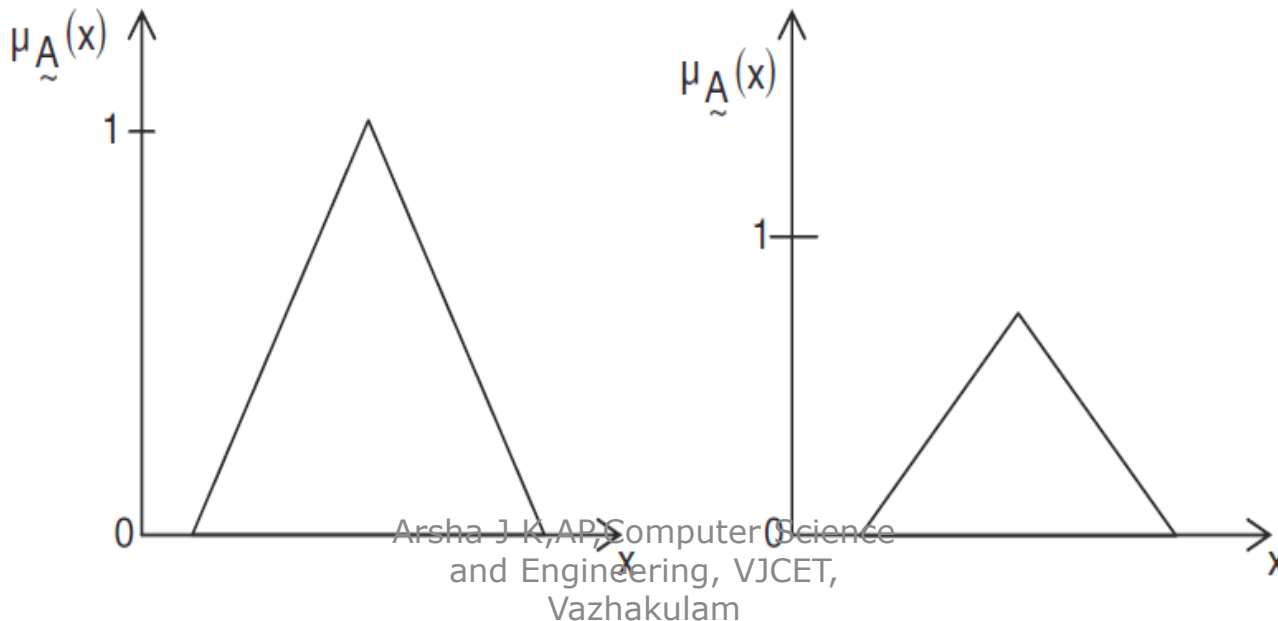
The boundary comprises those elements of x of the universe such that

$$0 < \mu_{A^{\sim}}(x) < 1$$

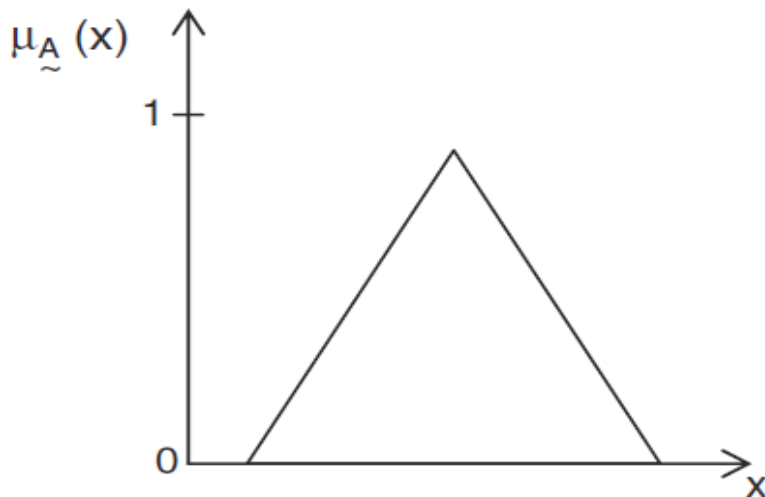
The boundary elements are those which possess partial membership in the fuzzy set $A^{\sim}$

OTHER TYPE OF FUZZY SETS

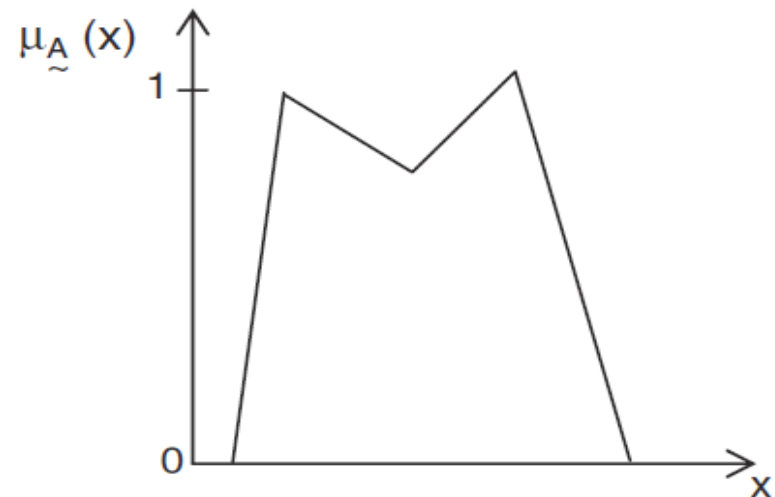
- A fuzzy set whose membership function has at least one element x in the universe whose membership value is unity is called **normal fuzzy set**
- The element for which membership is equal to 1 is called **prototypical elements**
- A fuzzy set wherein no membership function has its value equal to 1 is called **subnormal fuzzy set**



- A **convex fuzzy set** has a membership function whose membership value are strictly monotonically increasing or decreasing or strictly monotonically increasing than decreasing with increasing values for elements in the universe
- A fuzzy set possessing characteristics opposite that of convex fuzzy set is called **nonconvex fuzzy set**
 - That is membership values of membership function are not strictly monotonically increasing or decreasing or strictly monotonically increasing than decreasing



(a) Convex fuzzy set



(b) Nonconvex fuzzy set

When intersection is performed on two convex fuzzy sets, the intersected portion is also a convex fuzzy set.

This is shown in Fig.4.4.

The shaded portions show that the intersected portion is also a convex fuzzy set. The membership functions can have different shapes like triangle, trapezoidal, Gaussian, etc.

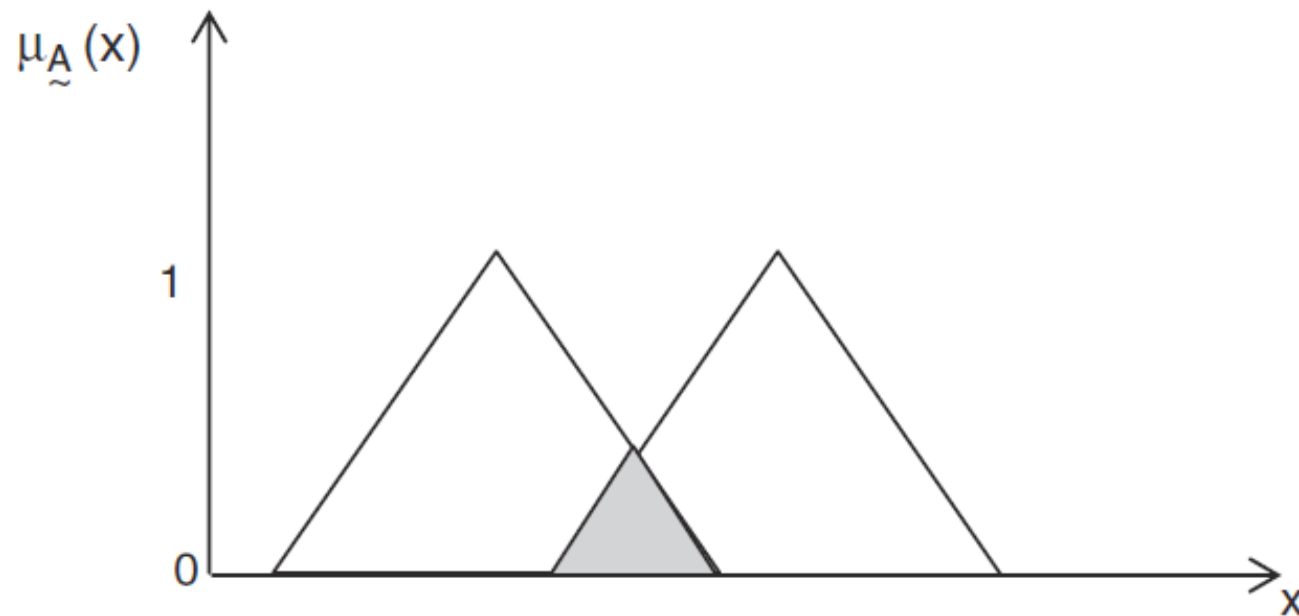


Fig. 4.4. Intersection of two convex sets

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The *crossover points* of a membership function are defined as the elements in the universe for which a particular fuzzy set $\tilde{A}$ has values equal to 0.5, that is, for which $\mu_{\tilde{A}}(x) = 0.5$.

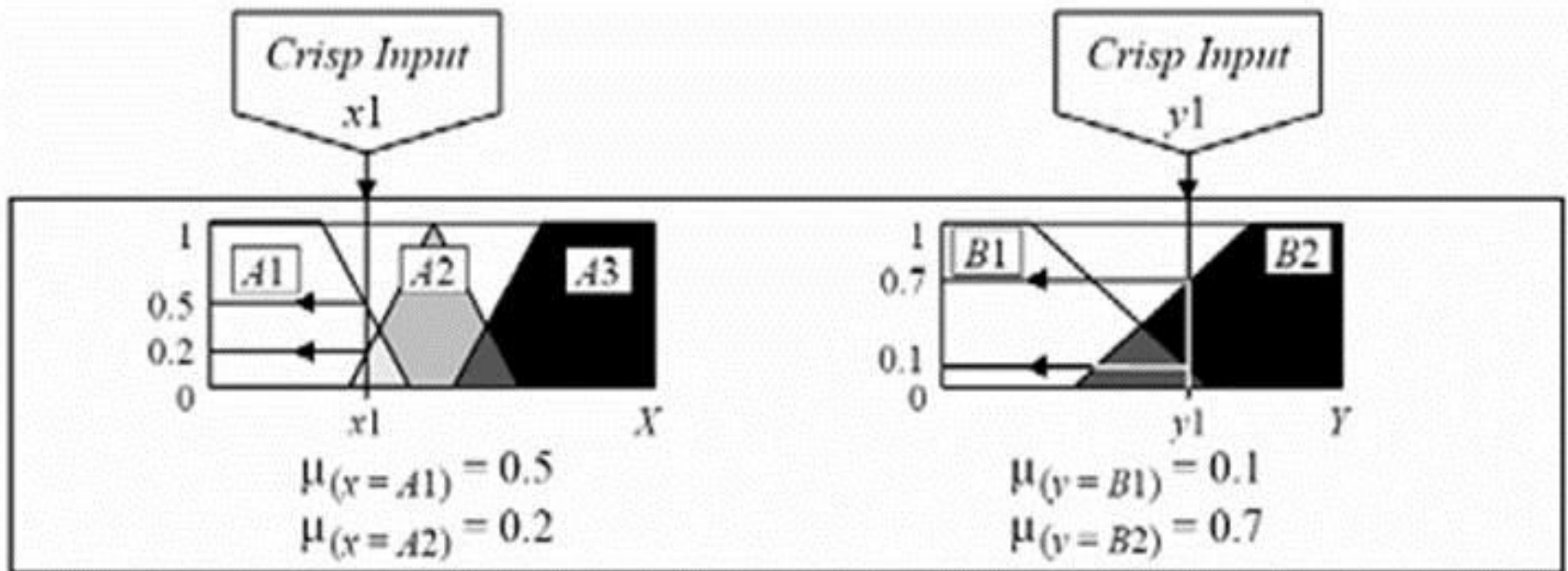
The *height* of a fuzzy set $\tilde{A}$ is the maximum value of the membership function, that is, $\text{hgt}(\tilde{A}) = \max\{\mu_{\tilde{A}}(x)\}$. If the $\text{hgt}(\tilde{A}) < 1$, the fuzzy set is said to be subnormal. The $\text{hgt}(\tilde{A})$ may be viewed as the degree of validity or credibility of information expressed by $\tilde{A}$ (Klir and Yuan, 1995).

If $\tilde{A}$ is a convex single-point normal fuzzy set defined on the real line, then $\tilde{A}$ is often termed a *fuzzy number*.

FUZZIFICATION

- **Fuzzifier converts a crisp input into a fuzzy variable.**
- **Definition of the membership functions must**
 - **reflects the designer's knowledge**
 - **provides smooth transition between member and nonmembers of a fuzzy set**
 - **simple to calculate**
- **Typical shapes of the membership function are Gaussian, trapezoidal and triangular.**

- Use crisp inputs from the user.
- Determine membership values for all the relevant classes (i.e., in right Universe of Discourse).



EXAMPLE - FUZZIFICATION

- **Assume we want to evaluate the health of a person based on his height and weight.**
- **The input variables are the crisp numbers of the person's height and weight.**
- **Fuzzification is a process by which the numbers are changes into linguistic words**

LINGUISTIC VARIABLE

- A **linguistic variable** associates words or sentences with a measure of **belief functions**, also called **membership function**.
- The set of values that it can take is called **term set**.
- Each value in the set is a **fuzzy variable** defined over a **base variable**.
- The base variable defines the **Universe of discourse** for all the fuzzy variables in the term set.

A **linguistic variable** is a quintuple $[X, T(X), U, G, M]$ where

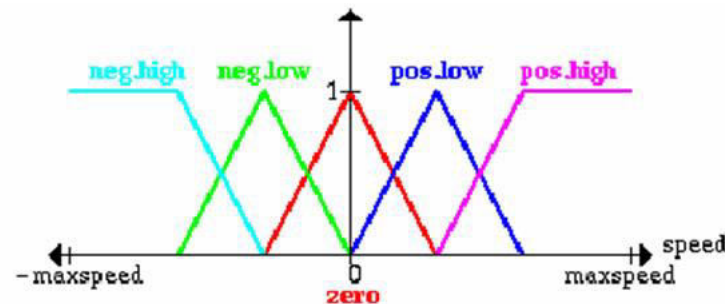
- X is the name of the variable,
- $T(X)$ is the term set, i.e. the set of names of linguistic values of X ,
- U is the universe of discourse,
- G is the grammar to generate the names and
- M is a set of semantic rules for associating each X with its meaning.

LINGUISTIC VARIABLE

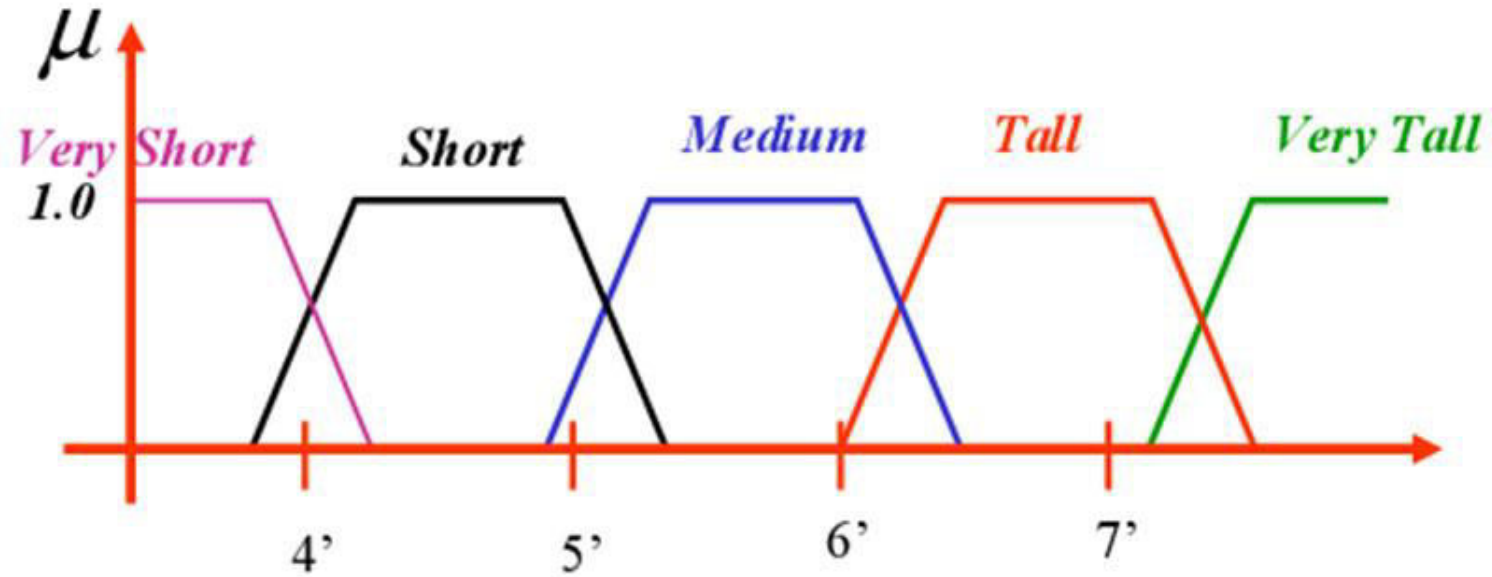
- Let x be a linguistic variable with the label "speed".
- Terms of x , which are fuzzy sets, could be "positive low", "negative high" from the term set T :

$$T = \{PositiveHigh, PositiveLow, NegativeLow, NegativeHigh, Zero\}$$

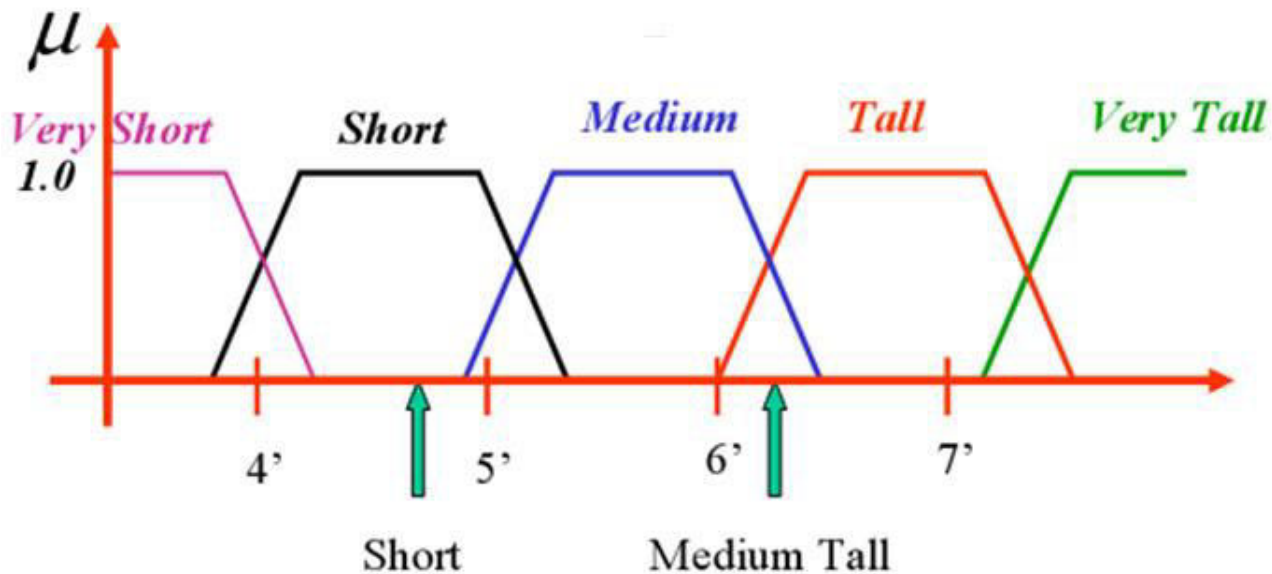
- Each term is a fuzzy variable defined on the base variable which might be the scale of all relevant velocities.



MEMBERSHIP FUNCTIONS



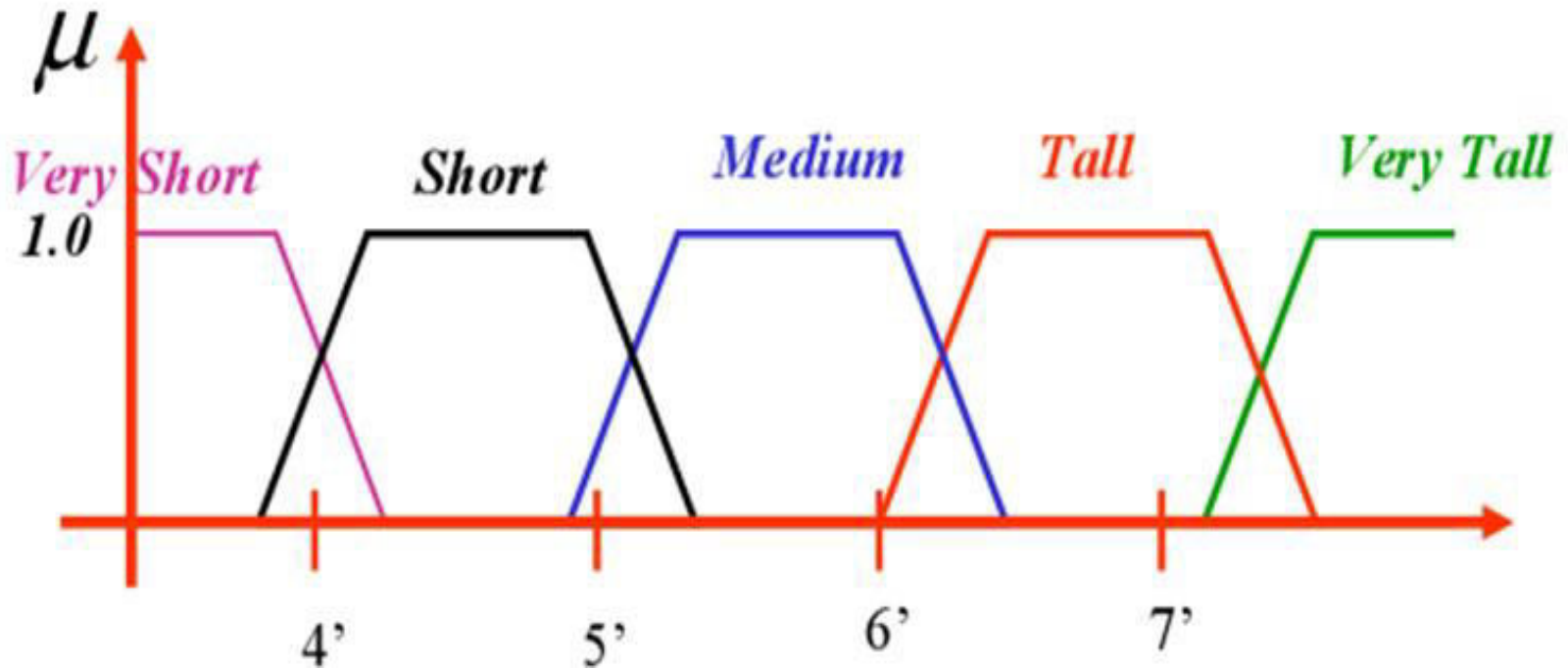
$$\mu = [\mu_{vs}, \mu_s, \mu_m, \mu_t, \mu_{vt}]$$



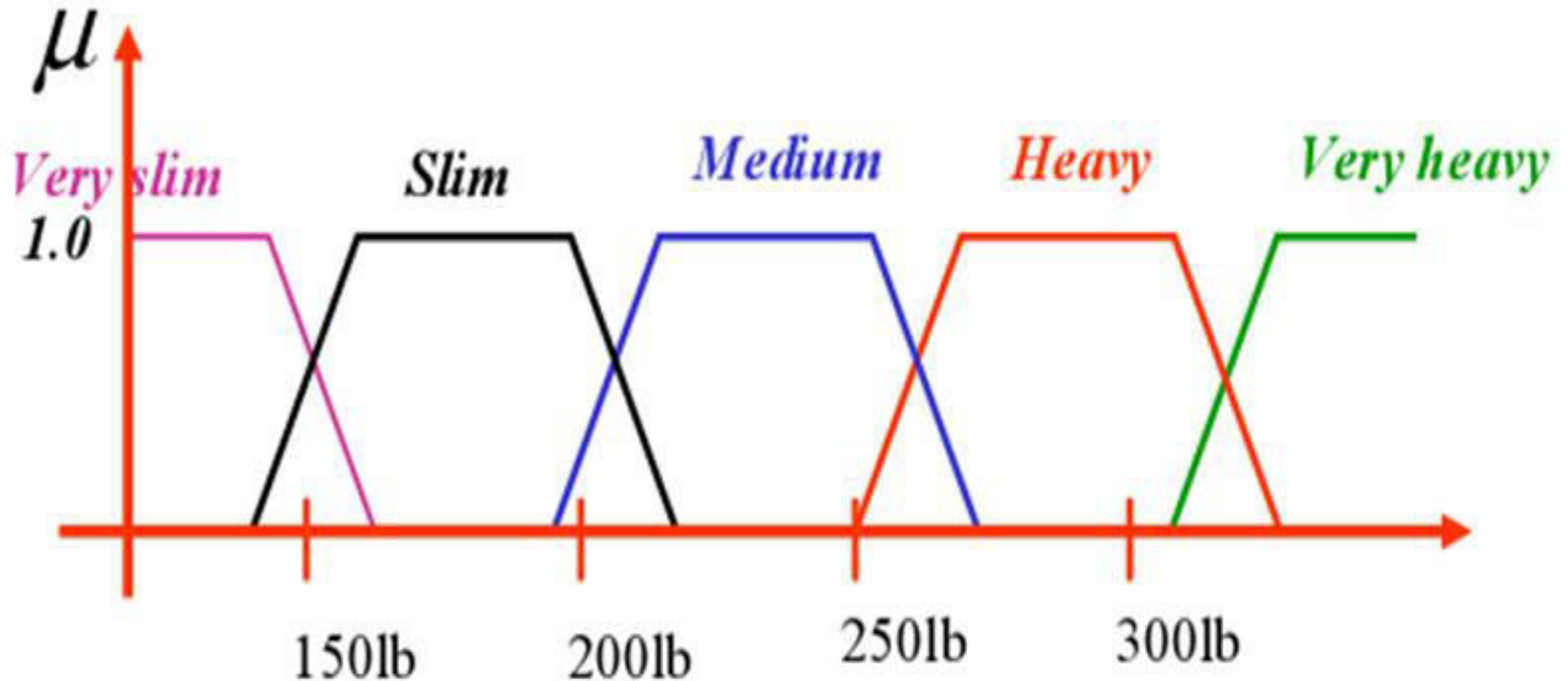
$$\mu = [0, 1, 0, 0, 0]$$

$$\mu = [0, 0, 0.5, 0.5, 0]$$

FUZZIFICATION OF HEIGHT



FUZZIFICATION OF WEIGHT



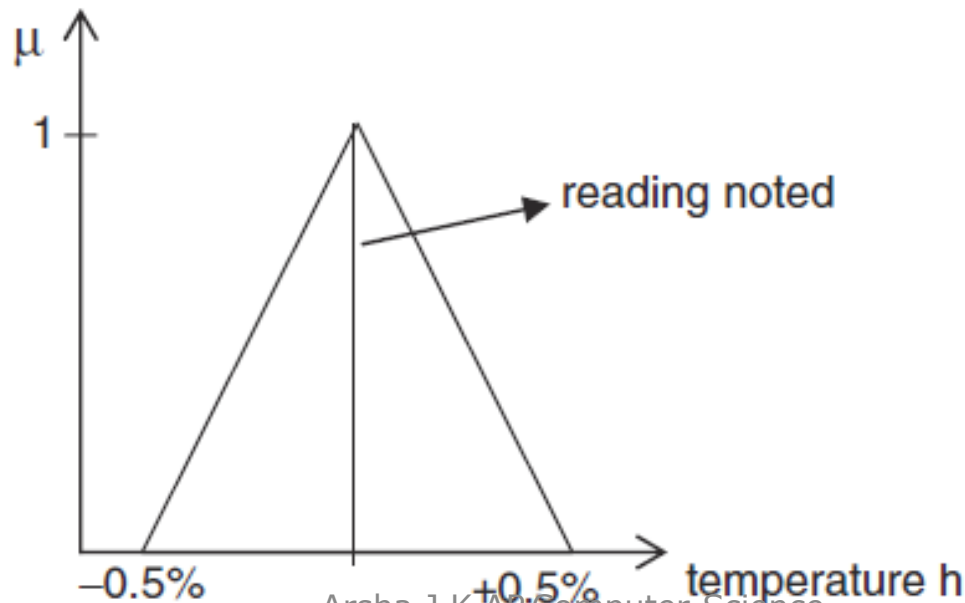
METHODS OF MEMBERSHIP VALUE ASSIGNMENT

The various methods of assigning membership values are:

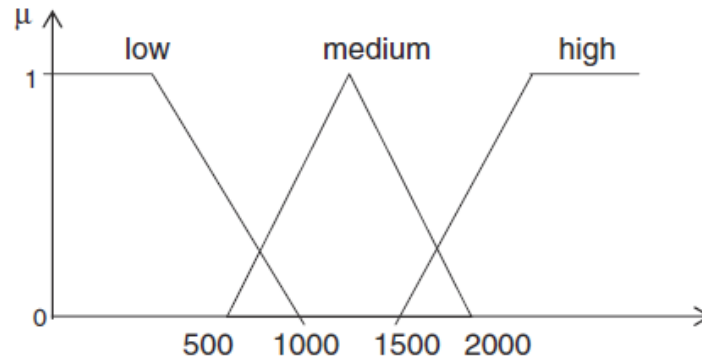
- Intuition,
- Inference,
- Rank ordering,
- Angular fuzzy sets,
- Neural networks,
- Genetic algorithm,
- Inductive reasoning.

Intuition

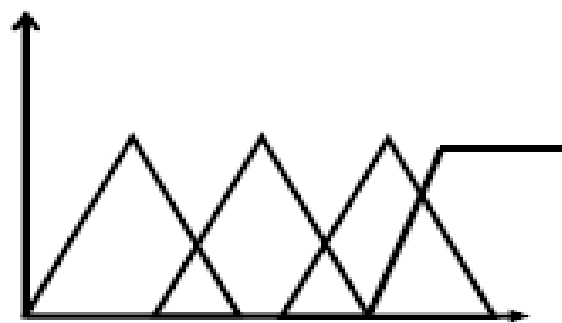
- Intuition is based on the human's own intelligence and understanding to develop the membership functions.
- The thorough knowledge of the problem has to be known, the knowledge regarding the linguistic variable should also be known.



Intuition



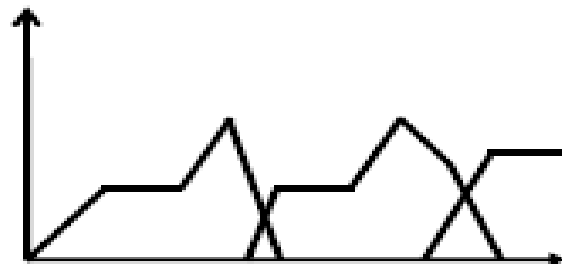
- Consider the speed of dc motor
- The curve represent the membership function corresponding to various fuzzy variable
- The range of speed is spitted into low , medium and high
- curves differentiate the ranges, said by humans.
- The placement of curves is approximate over the universe of discourse; the number of curves and the overlapping of curves is an important criteria to be considered while defining membership functions.



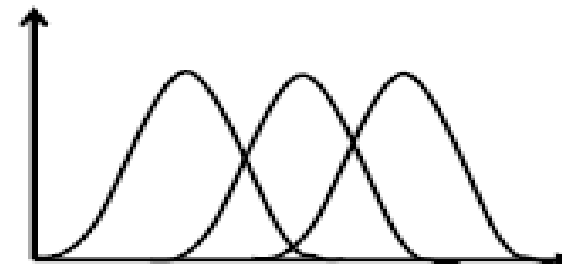
triangul



trapezoidal



piecewise linear



Gaussian

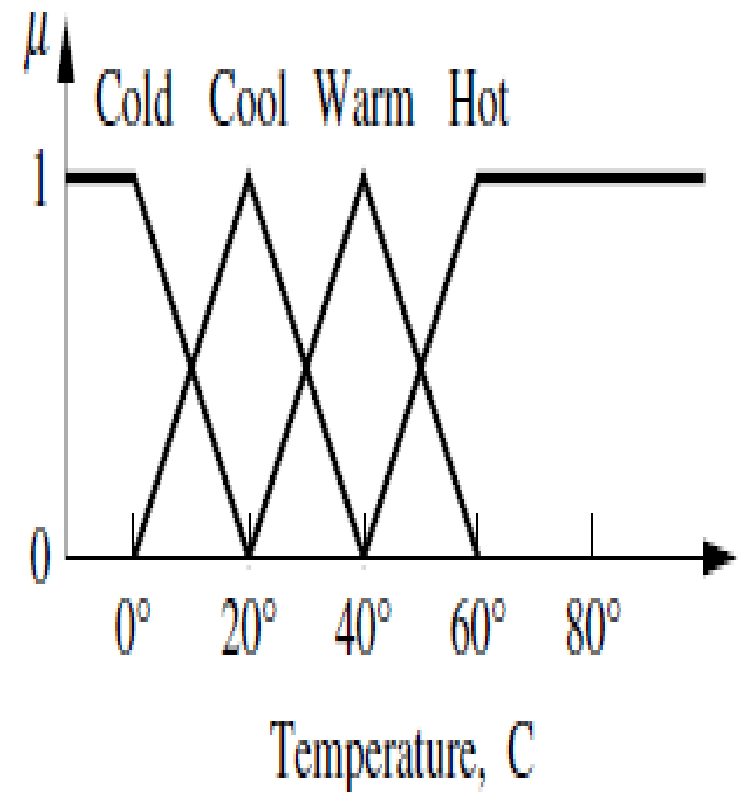


singleton

Figure 6.1 Different types of membership functions

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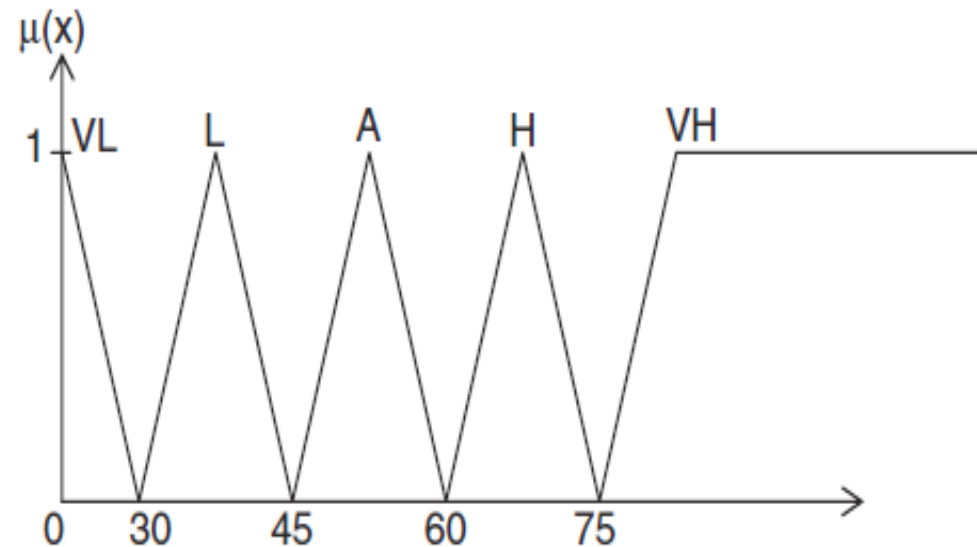
- As an example, consider the membership functions for the fuzzy variable temperature.
- Figure shows various shapes on the universe of temperature as measured in units of degrees Celsius.
- Each curve is a membership function corresponding to various fuzzy variables, such as very cold, cold, normal, hot, and very hot.
- For example, if the temperatures are referred to the range of human comfort we get one set of curves, and if they are referred to the range of safe operating temperatures for a steam turbine we get another set.



1. Using your own intuition and definitions of the universe of discourse, plot fuzzy membership functions for “weight of people.”

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- The universe of discourse is the weight of people. Let the weights be in “kg” – kilogram.
- Let the linguistic variables are:
 - Very light - $w \leq 30$
 - Light - $30 < w \leq 45$
 - Average - $45 < w \leq 60$
 - Heavy - $60 < w \leq 75$
 - Very heavy - $w > 75$

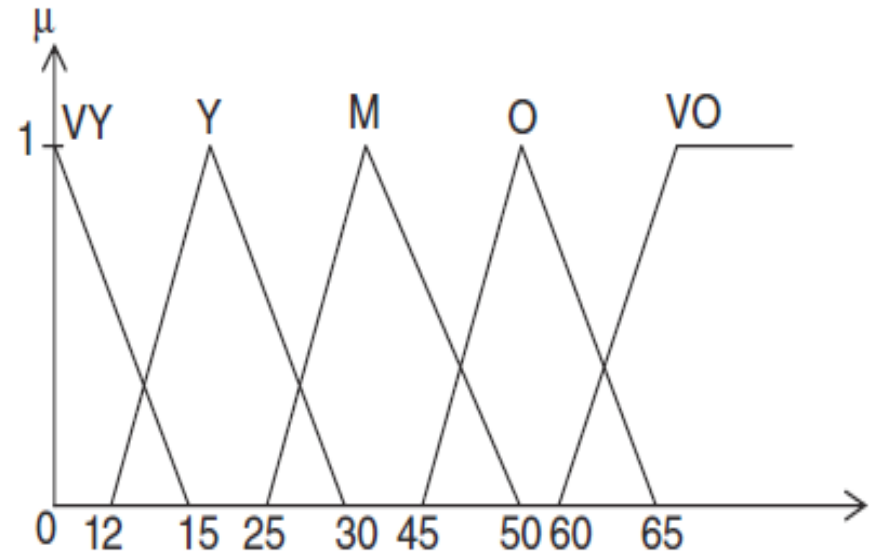


- Representing this using triangular membership function, as shown in

Using your own intuition, plot the fuzzy membership function for the age of people.

Using your own intuition, plot the fuzzy membership function for the age of people.

- The linguistic variables are defined as, let A denotes age in years.
 - (1) Very young (vy) – $A < 15$
 - (2) Young (y) – $12 \leq A < 30$
 - (3) Middle aged (m) – $25 \leq A < 50$
 - (4) Old (o) – $45 \leq A < 65$
 - (5) Very old (vo) – $60 < A$



- This is represented using triangular membership, as shown in

Compare medium wave and short wave receivers according to their frequency range. Plot the membership function using intuition. The linguistic variable are defined based on the following

Medium wave receivers: frequency lesser than 10^6 Hz

Short wave receivers : Frequency greater than 10^6 Hz

Solution

Let the frequency range of receivers be the universe of discourse. The linguistic variables are following

Medium wave receivers:
frequency lesser than 10^6 Hz

Short wave receivers :
Frequency greater than 10^6 Hz

Using your own intuition and your own definition of the universe of discourse plot fuzzy membership functions to the following variables:

(1) Height of liquid in a tank

- **(a) Very full**
- **(b) Full**
- **(c) Medium**
- **(d) Small**
- **(e) Very small**

(2) Race of people

- **(a) Very white**
- **(b) White**
- **(c) Moderate**
- **(d) Black**
- **(e) Very black**

- Using your own intuition, develop fuzzy membership functions for the fuzzy number 3, using the following shapes:
- (a) right angle triangle,
- (b) quadrilateral,
- (c) Gaussian function,
- (d) trapezoid, and
- (e) isosceles triangle.

- Using intuition, assign the membership functions for (a) population of people, (b) employment strategy, and (c) usage of library.

- Using your own intuition, develop fuzzy membership functions on the real line for the fuzzy number 4, using the following function shapes:
- (1) Symmetric triangle
- (2) Trapezoid
- (3) Gaussian function

- Using your own intuition, develop fuzzy number “approximately 4 or approximately 8” using the following function shapes:
- (1) Symmetric triangle
- (2) Trapezoids
- (3) Gaussian functions.

INFERENCE

- This method involves the knowledge to perform deductive reasoning.
- The membership function is formed from the facts known and knowledge.
- That is, we wish to deduce or infer a conclusion, given a body of facts and knowledge

- Let us use inference method for the identification of the triangle.
- Let U be universe of triangles and A , B , and C be the inner angles of the triangles.
- Also $A \geq B \geq C \geq 0$. Therefore the universe is given by:

$$U = \{(A, B, C) \mid A \geq B \geq C \geq 0; A + B + C = 180^\circ\}$$

$I_{\sim}$	Approximate isosceles triangle
$R_{\sim}$	Approximate right triangle
$IR_{\sim}$	Approximate isosceles <i>and</i> right triangle
$E_{\sim}$	Approximate equilateral triangle
$T_{\sim}$	Other triangles

- For the approximate isosceles triangle we have the following algorithm for the membership, again for the situation of

$$A \geq B \geq C \geq 0 \text{ and } A + B + C = 180^\circ$$

$$\mu_{\sim I}(A, B, C) = 1 - \frac{1}{60^\circ} \min(A - B, B - C)$$

- So, for example, if $A = B$ or $B = C$, the membership value in the approximate isosceles triangle is $\mu_{\sim I} = 1$; if $A = 120^\circ$, $B = 60^\circ$, and $C = 0^\circ$, then $\mu_{\sim I} = 0$.

- For a fuzzy right triangle, we have

$$\mu_{\tilde{R}}(A, B, C) = 1 - \frac{1}{90^\circ} |A - 90^\circ|$$

- For instance, when $A = 90^\circ$, the membership value in the fuzzy right triangle, $\mu_{\tilde{R}} = 1$,
- or when $A = 180^\circ$, this membership vanishes, i.e., $\mu_{\tilde{R}} = 0$.

- For the case of an approximate isosceles *and* right triangle (there is only one of these in the crisp domain), we can find this membership function by taking the logical intersection (*and* operator) of the isosceles *and* right triangle membership functions, or

$$\mathbb{I}\mathbb{R} = \mathbb{I} \cap \mathbb{R}$$

$$\mu_{\mathbb{I}\mathbb{R}}(A, B, C) = \min[\mu_{\mathbb{I}}(A, B, C), \mu_{\mathbb{R}}(A, B, C)]$$

$$= 1 - \max \left[\frac{1}{60^\circ} \min(A - B, B - C), \frac{1}{90^\circ} |A - 90^\circ| \right]$$

For the case of a fuzzy equilateral triangle, the membership function is given by

$$\mu_{E\sim}(A, B, C) = 1 - \frac{1}{180^\circ}(A - C)$$

For example, when $A = B = C$, the membership value is $\mu_{E\sim}(A, B, C) = 1$; when $A = 180^\circ$, the membership value vanishes, or $\mu_{E\sim} = 0$.

- Finally, for the set of ‘‘all other triangles’’(all triangular shapes other than $I_{\sim}$, $R_{\sim}$, and $E_{\sim}$) we simply invoke the complement of the logical union of the three previous cases (or, from De Morgan’s principles the intersection of the complements of the triangular shapes),

$$T_{\sim} = \overline{(I_{\sim} \cup R_{\sim} \cup E_{\sim})} = \bar{I}_{\sim} \cap \bar{R}_{\sim} \cap \bar{E}_{\sim}$$

which results in

$$\begin{aligned} \mu_{T_{\sim}}(A, B, C) &= \min\{1 - \mu_{I_{\sim}}(A, B, C), 1 - \mu_{E_{\sim}}(A, B, C), 1 - \mu_{R_{\sim}}(A, B, C)\} \\ &= \frac{1}{180^\circ} \min\{3(A - B), 3(B - C), 2|A - 90^\circ|, A - C\} \end{aligned}$$

Example 6.1 [Ross, 1995]. Define a specific triangle, as shown in Fig. 6.2, with these three ordered angles:

$$\{X : A = 85^\circ \geq B = 50^\circ \geq C = 45^\circ, \text{ where } A + B + C = 180^\circ\}$$

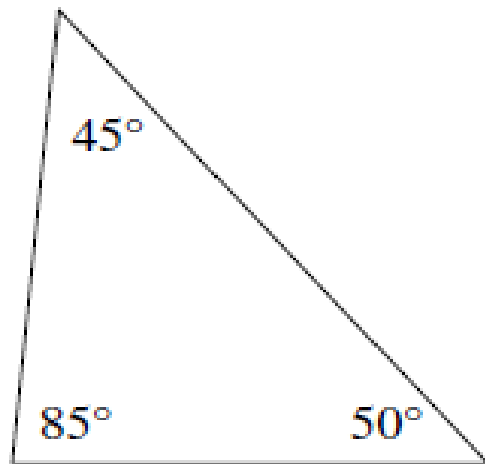


FIGURE 6.2
A specific triangle.

- Using the inference method, find the membership values of the triangular shapes for each of the following triangles:
- (a) 60° , 40° , 80° ,
- (b) 45° , 65° , 70° , and
- (c) 75° , 55° , 50° .

- Using the Inference approach outlined in this chapter find the membership values for each of the triangular shapes ($I^\sim$, $R^\sim$, $IR^\sim$, $E^\sim$, $R^\sim$) for each of the following
- (1) 80° , 75° , 25° ,
- (2) 60° , 75° , 45° ,
- (3) 50° , 75° , 55° , and
- (4) 45° , 45° , 90° .

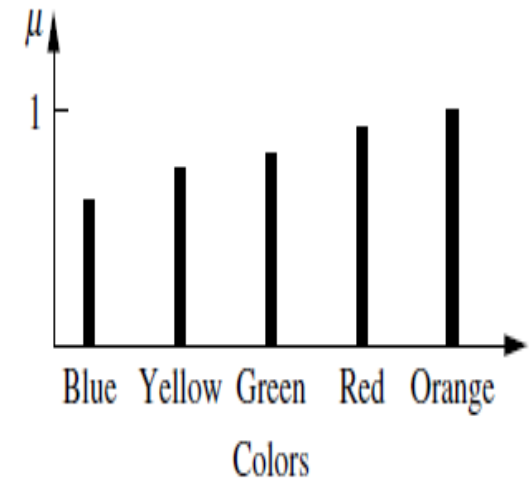
RANK ORDERING

- The polling concept is used to assign membership values by rank ordering process.
- Preferences are above for pair wise comparisons and from this the ordering of the membership is done.

- Suppose 1000 people respond to a questionnaire about their pair wise preferences among five colors, $X = \{\text{red, orange, yellow, green, blue}\}$.
- Define a fuzzy set as $A_{\sim}$ on the universe of colors ‘‘best color.’’ Table 6.1 is a summary of the opinion survey.
- In this table, for example, out of 1000 people 517 preferred the color red to the color orange, 841 preferred the color orange to the color yellow, etc.

TABLE 6.1
Example in Rank Ordering

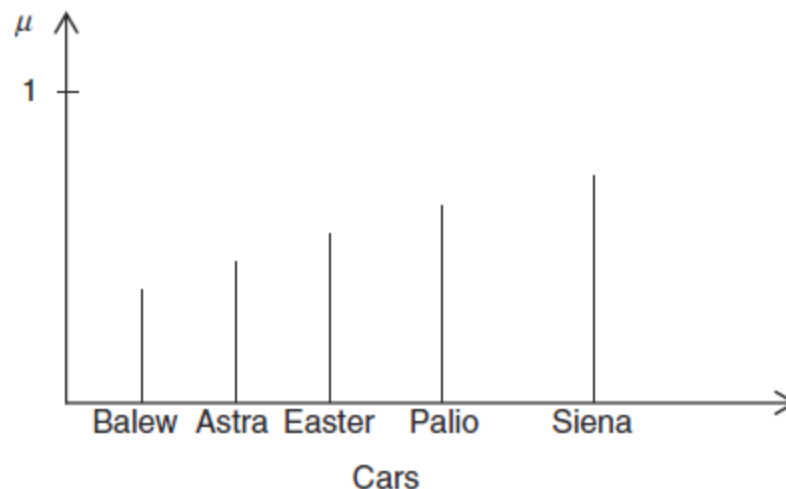
	Number who preferred					Total	Percentage	Rank order
	Red	Orange	Yellow	Green	Blue			
Red	–	517	525	545	661	2248	22.5	2
Orange	483	–	841	477	576	2377	23.8	1
Yellow	475	159	–	534	614	1782	17.8	4
Green	455	523	466	–	643	2087	20.9	3
Blue	339	424	386	357	–	1506	15	5
Total						10,000		



- Suppose 1,000 people responds to a questionnaire about the pair wise preference among five cars, $x = \{\text{Palio, Siena, Astra, Easter, Baleno}\}$.
- Define a fuzzy set as $A \sim$ on the universe of cars, “best cars”.

Table 4.1. Pairwise preferences among five cars between 1000 people

	Number who preferred					Total	Percentage	Rank order
	Palio	Siena	Astra	Easter	Baleno			
Palio	–	515	545	523	671	2,254	22.5	2
Siena	481	–	475	845	580	2,381	23.8	1
Astra	469	624	–	141	536	1,770	17.7	4
Easter	457	530	470	–	649	2,114	21.1	3
Baleno	265	425	402	389	–	1,481	14.8	5
Total						10,000		



- The following data were determined by the pairwise comparison of work preferences of 100 people. When it was compared with Software (S), 69 of persons polled preferred Hardware (H), 45 of them preferred Educational (E), 55 of them preferred Business (B) and 25 preferred Textile (T). When it was compared with hardware (H), the preferences was 58-S, 45-E, 60-B, 30-T. When it was compared with educational, 39-S, 56-H, 34-B, 25-T. When it was compared business, the preferences was 52-S, 49-H, 38-E, 20-T. When it was compared with textile, the preferences was 69-S, 65-H, 44-E, 40-B. Using rank ordering, plot the membership function for the “ m o s t p r e f e r r e d w o r k . ”

SUMMARY

The contents discussed in this chapter include:

- Membership functions – Definition and properties.
- Linguistic variables.
- Fuzzification process.

FUZZINESS vs PROBABILITY

- When first exposed to fuzzy logic, humans associate **membership functions** with **density functions**.
- This is not so, since:
 - Probability density is an **abstraction from empirical frequency**.
⇒ an aggregate property.
 - **how often** events occur in different ways.
 - ways that are quite **crisp and mutually exclusive** after occurrence.
 - Fuzzy relations, by contrast, **are properties of single events** that **are always there**, and not different from occurrence to occurrence.

CHAPTER 10

DEFUZZIFICATION

DEFUZZIFICATION

- Defuzzification is a mapping process from a space of fuzzy control actions defined over an output universe of discourse into a space of crisp (nonfuzzy) control actions.
- Defuzzification is a process of converting output fuzzy variable into a unique number.
- Defuzzification process has the capability to reduce a fuzzy set into a crisp single-valued quantity or into a crisp set; to convert a fuzzy matrix into a crisp matrix; or to convert a fuzzy number into a crisp number.

LAMBDA CUT FOR FUZZY SETS

Consider a fuzzy set $\underline{A}$. The set A_λ ($0 < \lambda < 1$) called the lambda (λ)-cut (or alpha [α]-cut) set is a crisp set of the fuzzy set and is defined as follows:

$$A_\lambda = \{x | \mu_{\underline{A}}(x) \geq \lambda\}; \quad \lambda \in [0, 1]$$

The properties of λ -cut sets are as follows:

1. $(\underline{A} \cup \underline{B})_\lambda = A_\lambda \cup B_\lambda$
2. $(\underline{A} \cap \underline{B})_\lambda = A_\lambda \cap B_\lambda$
3. $(\bar{\underline{A}})_\lambda \neq (\bar{A}_\lambda)$ except when $\lambda = 0.5$
4. For any $\lambda \leq \beta$, where $0 \leq \beta \leq 1$, it is true that $A_\beta \subseteq A_\lambda$, where $A_0 = X$.

LAMBDA CUT FOR FUZZY RELATIONS

Let $\underline{R}$ be a fuzzy relation where each row of the relational matrix is considered a fuzzy set. The j th row in a fuzzy relation matrix $\underline{R}$ denotes a discrete membership function for a fuzzy set $\underline{R}_j$. A fuzzy relation can be converted into a crisp relation in the following manner:

$$R_\lambda = \{(x, y) | \mu_{\underline{R}}(x, y) \geq \lambda\}$$

where R_λ is a λ -cut relation of the fuzzy relation $\underline{R}$.

For two fuzzy relations $\underline{R}$ and $\underline{S}$ the following properties should hold:

1. $(\underline{R} \cup \underline{S})_\lambda = R_\lambda \cup S_\lambda$
2. $(\underline{R} \cap \underline{S})_\lambda = R_\lambda \cap S_\lambda$
3. $(\tilde{\underline{R}})_\lambda \neq (\tilde{R}_\lambda)$ except when $\lambda = 0.5$
4. For any $\lambda \leq \beta$, where $0 \leq \beta \leq 1$, it is true that $R_\beta \subseteq R_\lambda$.

Example 5.1. Two fuzzy sets $\tilde{P}$ and $\tilde{Q}$ are defined on x as follows:

$\mu(x_1)$	x_1	x_2	x_3	x_4	x_5
$\tilde{P}$	0.1	0.2	0.7	0.5	0.4
$\tilde{Q}$	0.9	0.6	0.3	0.2	0.8

Find the following λ cut sets

- (a) $\left(\overline{\tilde{P}}\right)_{0.2}$ (b) $(\tilde{Q})_{0.3}$ (c) $\left(\tilde{P} \cup \tilde{Q}\right)_{0.5}$ (d) $\left(\tilde{P} \cap \tilde{Q}\right)_{0.4}$
 (e) $\left(\tilde{Q} \cup \overline{\tilde{P}}\right)_{0.8}$ (f) $\left(\tilde{P} \cup \overline{\tilde{P}}\right)_{0.2}.$

Solution. Given

$$P_{\sim} = \left\{ \frac{0.1}{x_1} + \frac{0.2}{x_2} + \frac{0.7}{x_3} + \frac{0.5}{x_4} + \frac{0.4}{x_5} \right\},$$

$$Q_{\sim} = \left\{ \frac{0.9}{x_1} + \frac{0.6}{x_2} + \frac{0.3}{x_3} + \frac{0.2}{x_4} + \frac{0.8}{x_5} \right\}.$$

Finding

$$\overline{P}_{\sim} = \left\{ \frac{0.9}{x_1} + \frac{0.8}{x_2} + \frac{0.3}{x_3} + \frac{0.5}{x_4} + \frac{0.6}{x_5} \right\},$$

$$\overline{Q}_{\sim} = \left\{ \frac{0.1}{x_1} + \frac{0.4}{x_2} + \frac{0.7}{x_3} + \frac{0.8}{x_4} + \frac{0.2}{x_5} \right\}.$$

$$(a) \quad \left(\overline{P} \right)_{\sim 0.2} = \left\{ \frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{1}{x_4} + \frac{1}{x_5} \right\},$$

$$(b) \quad \left(\overline{Q} \right)_{\sim 0.3} = \left\{ \frac{0}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{1}{x_4} + \frac{0}{x_5} \right\},$$

$$(c) \quad \left(P \cup Q \right)_{\sim} = \left\{ \frac{0.9}{x_1} + \frac{0.6}{x_2} + \frac{0.7}{x_3} + \frac{0.5}{x_4} + \frac{0.8}{x_5} \right\},$$

$$\left(P \cup Q \right)_{\sim 0.6} = \left\{ \frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{0}{x_4} + \frac{1}{x_5} \right\},$$

$$(d) \quad \left(P \cup \overline{P} \right)_{\sim} = \left\{ \frac{0.9}{x_1} + \frac{0.8}{x_2} + \frac{0.7}{x_3} + \frac{0.5}{x_4} + \frac{0.6}{x_5} \right\},$$

$$\left(P \cup \overline{P} \right)_{\sim 0.8} = \left\{ \frac{1}{x_1} + \frac{1}{x_2} + \frac{0}{x_3} + \frac{0}{x_4} + \frac{0}{x_5} \right\}$$

$$\begin{aligned}
 \text{(e)} \quad & \begin{pmatrix} P \cap Q \\ \sim \quad \sim \end{pmatrix} = \left\{ \frac{0.9}{x_1} + \frac{0.8}{x_2} + \frac{0.7}{x_3} + \frac{0.5}{x_4} + \frac{0.6}{x_5} \right\}, \\
 & \begin{pmatrix} P \cap Q \\ \sim \quad \sim \end{pmatrix}_{0.4} = \left\{ \frac{0}{x_1} + \frac{0}{x_2} + \frac{0}{x_3} + \frac{0}{x_4} + \frac{1}{x_5} \right\}, \\
 \text{(f)} \quad & \begin{pmatrix} P \cap \overline{P} \\ \sim \quad \sim \end{pmatrix} = \left\{ \frac{0.1}{x_1} + \frac{0.2}{x_2} + \frac{0.3}{x_3} + \frac{0.5}{x_4} + \frac{0.4}{x_5} \right\}, \\
 & \begin{pmatrix} P \cap \overline{P} \\ \sim \quad \sim \end{pmatrix}_{0.8} = \left\{ \frac{0}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{1}{x_4} + \frac{1}{x_5} \right\}.
 \end{aligned}$$

Example 5.2. Given three fuzzy sets:

$$A_{\sim} = \left\{ \frac{0.9}{x_1} + \frac{0.5}{x_2} + \frac{0.2}{x_3} + \frac{0.3}{x_4} \right\},$$

$$B_{\sim} = \left\{ \frac{0.2}{x_1} + \frac{1.0}{x_2} + \frac{0.8}{x_3} + \frac{0.4}{x_4} \right\},$$

$$C_{\sim} = \left\{ \frac{0.1}{x_1} + \frac{0.7}{x_2} + \frac{0.5}{x_3} + \frac{0.6}{x_4} \right\}.$$

Find $A_{0.6}$, $B_{1.0}$, $C_{0.3}$, $\overline{A}_{\sim_{0.2}}$, $\overline{B}_{\sim_{0.8}}$, $\overline{C}_{\sim_{0.5}}$.

$$(a) \quad A_{0.6} = \left\{ \frac{1}{x_1} + \frac{0}{x_2} + \frac{0}{x_3} + \frac{0}{x_4} \right\},$$

$$(b) \quad B_{1.0} = \left\{ \frac{0}{x_1} + \frac{1}{x_2} + \frac{0}{x_3} + \frac{0}{x_4} \right\},$$

$$(c) \quad C_{0.3} = \left\{ \frac{0}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{1}{x_4} \right\},$$

$$(d) \quad \begin{aligned} \overline{A}_{\sim} &= \left\{ \frac{0.1}{x_1} + \frac{0.5}{x_2} + \frac{0.8}{x_3} + \frac{0.7}{x_4} \right\}, \\ \overline{A}_{\sim 0.2} &= \left\{ \frac{0}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{1}{x_4} \right\}, \end{aligned}$$

$$(e) \quad \begin{aligned} \overline{B}_{\sim} &= \left\{ \frac{0.8}{x_1} + \frac{0.0}{x_2} + \frac{0.2}{x_3} + \frac{0.6}{x_4} \right\}, \\ \overline{B}_{\sim 0.8} &= \left\{ \frac{1}{x_1} + \frac{0}{x_2} + \frac{0}{x_3} + \frac{0.9}{x_4} \right\}, \end{aligned}$$

$$(f) \quad \begin{aligned} \overline{C}_{\sim} &= \left\{ \frac{0.9}{x_1} + \frac{0.3}{x_2} + \frac{0.5}{x_3} + \frac{0.4}{x_4} \right\}, \\ \overline{C}_{\sim 0.5} &= \left\{ \frac{1}{x_1} + \frac{0}{x_2} + \frac{1}{x_3} + \frac{0}{x_4} \right\}. \end{aligned}$$

Example 5.3. The fuzzy sets $\tilde{A}$ and $\tilde{B}$ are defined as universe, $x = \{0, 1, 2, 3\}$, with the following membership fractions:

$$\mu_{\tilde{A}}(x) = \frac{2}{x+3},$$

$$\mu_{\tilde{B}}(x) = \frac{4x}{x+5}.$$

Define the intervals along x -axis corresponding to the λ cut sets for each fuzzy set $\tilde{A}$ and $\tilde{B}$ for following values of λ . $\lambda = 0.2, 0.5, 0.6$.

Solution.

$$x = \{0, 1, 2, 3\}$$

X	0	1	2	3
$\mu_{A\sim}(x) = 2/x + 3$	$2/3$	$2/4$	$2/5$	$2/6$
$\mu_{B\sim}(x) = 4x/2(x+5)$	0	$4/12$	$8/14$	$12/16$

Therefore the values are:

X	0	1	2	3
$\mu_{A\sim}(x)$	0.67	0.5	0.4	0.33
$\mu_{B\sim}(x)$	0	0.33	0.57	0.75

(a) When $\lambda = 0.2$

$$A_{0.2} = \{0, 1, 2, 3\},$$

$$A_{0.2} = \{1, 2, 3\}.$$

(b) When $\lambda = 0.5$

$$A_{0.5} = \{0, 1\},$$

$$B_{0.5} = \{2, 3\}.$$

(c) When $\lambda = 0.6$

$$A_{0.6} = \{0\},$$

$$A_{0.6} = \{3\}.$$

Example 5.4. For the fuzzy relation

$$R_{\sim} = \begin{bmatrix} 1 & 0.2 & 0.3 \\ 0.5 & 0.9 & 0.6 \\ 0.4 & 0.8 & 0.7 \end{bmatrix},$$

find the λ cut relations for the following values of $\lambda = 0^+, 0.2, 0.9, 0.5$.

Solution. Given,

$$R_{\sim} = \begin{bmatrix} 1 & 0.2 & 0.3 \\ 0.5 & 0.9 & 0.6 \\ 0.4 & 0.8 & 0.7 \end{bmatrix}.$$

(a) $\lambda = 0^+$,

$$R_{0^+} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

(b) $\lambda = 0.2$,

$$R_{0.2} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

(c) $\lambda = 0.9$,

$$R_{0.9} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

(d) $\lambda = 0.5$,

$$R_{0.5} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Example 5.5. For the given fuzzy relation

$$R = \begin{bmatrix} 0.2 & 0.5 & 0.47 & 1 & 0.9 \\ 0.3 & 0.5 & 0.6 & 1 & 0.8 \\ 0.4 & 0.6 & 0.4 & 0.5 & 0.3 \\ 0.9 & 1 & 0.3 & 0.3 & 0.2 \end{bmatrix},$$

find the cut λ cut relation for the following values of $\lambda = 0.4, 0.7, 0.8$.

Solution. (a) $\lambda = 0.4$,

$$R_{0.4} = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}.$$

(b) $\lambda = 0.7$,

$$R_{0.7} = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}.$$

(c) $\lambda = 0.8$,

$$R_{0.8} = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}.$$

1. Determine crisp λ cut relation for $\lambda = 0.2$; for $j = 0, 1, \dots, 10$ for the following fuzzy relation matrix $\tilde{R}$:

$$\tilde{R} = \begin{bmatrix} 0.3 & 0.8 & 0.7 & 0.9 \\ 1 & 0.7 & 0.6 & 0.2 \\ 0.1 & 0.7 & 1 & 0.9 \\ 0.5 & 0.6 & 0.2 & 0.5 \end{bmatrix}.$$

2. The fuzzy set $\tilde{A}$, $\tilde{B}$, $\tilde{C}$ are all defined on the universe $X = [0, 5]$ with the following membership functions:

$$\mu_{\tilde{A}}(x) = \frac{1}{1 + 5(x - 5)^2},$$

$$\mu_{\tilde{B}}(x) = 2^{-x},$$

$$\mu_{\tilde{C}}(x) = \frac{2x}{x + 5}.$$

- (a) Sketch the membership functions
- (b) Define the intervals along x -axis corresponding to the λ cut sets for each of the fuzzy sets $\tilde{A}$, $\tilde{B}$, $\tilde{C}$ for the following values of λ .

$$\lambda = 0.2, \lambda = 0.4, \lambda = 0.7, \lambda = 0.9$$

5. Two fuzzy sets $\underset{\sim}{A}$ and $\underset{\sim}{B}$ both defined on x are as follows:

$$\underset{\sim}{A} = \left\{ \frac{0.1}{x_1} + \frac{0.6}{x_2} + \frac{0.4}{x_3} + \frac{0.7}{x_4} + \frac{0.5}{x_5} + \frac{0.2}{x_6} \right\},$$

$$\underset{\sim}{B} = \left\{ \frac{0.8}{x_1} + \frac{0.6}{x_2} + \frac{0.3}{x_3} + \frac{0.2}{x_4} + \frac{0.6}{x_5} + \frac{0}{x_6} \right\}.$$

Find

(a) $\left(\underset{\sim}{\bar{A}} \right)_{0.5}$

(b) $\left(\underset{\sim}{B} \right)_{0.3}$

(c) $\left(\underset{\sim}{A} \cup \underset{\sim}{\bar{A}} \right)_{0.5}$

(d) $\left(\underset{\sim}{A} \cup \underset{\sim}{B} \right)_{0.4}$

(e) $\left(\overline{\underset{\sim}{A} \cap \underset{\sim}{B}} \right)_{0.6}$

(f) $\left(\underset{\sim}{A} \cap \underset{\sim}{B} \right)$

For fuzzy relation R find λ cut relations for the following values of λ

$$R = \begin{bmatrix} 0.4 & 0.3 & 0.7 & 0.5 \\ 0.6 & 0.2 & 0.1 & 1 \\ 0.9 & 0.8 & 0.5 & 0.6 \\ 0.7 & 0.4 & 0.3 & 0.2 \end{bmatrix}.$$

- | | | |
|---------------------|---------------------|---------------------|
| (a) $\lambda = 0^+$ | (c) $\lambda = 0.4$ | (e) $\lambda = 0.3$ |
| (b) $\lambda = 0.2$ | (d) $\lambda = 0.7$ | (f) $\lambda = 0.6$ |

6. For fuzzy relation $\underset{\sim}{A}$ and $\underset{\sim}{E}$ determine λ cut relations for the following values of λ :
- (a) $\lambda = 0^+$ (b) $\lambda = 0.5$ (c) $\lambda = 0.9$

$$\underset{\sim}{A} = \begin{bmatrix} 0.8 & 1 & 0.5 & 0.3 & 0.1 & 0 \\ 0.2 & 0.3 & 0.5 & 0.7 & 0.1 & 0.2 \\ 0.1 & 0.2 & 0.4 & 0.8 & 0.7 & 0.1 \\ 0.2 & 0.1 & 0.4 & 0.7 & 0.7 & 0.3 \end{bmatrix},$$

$$\underset{\sim}{E} = \begin{bmatrix} 0.8 & 0.7 & 0.4 & 0.1 & 0 \\ 0.6 & 0.5 & 0.3 & 0.2 & 0.1 \\ 0.9 & 0.6 & 0.7 & 0.4 & 0.3 \\ 0.2 & 0.4 & 0.5 & 0.9 & 0.6 \\ 0.1 & 0.4 & 0.3 & 0.6 & 0.9 \\ 0.1 & 0 & 1 & 0.8 & 0.7 \end{bmatrix}.$$

7. Determine the λ cut sets for the six set operation for two fuzzy set R and S using $\lambda = 0.2$ and 0.8 :

$$A_{\sim} = \left\{ \frac{0.1}{20} + \frac{0.6}{40} + \frac{0.4}{60} + \frac{0.3}{80} + \frac{0.9}{100} \right\},$$

$$B_{\sim} = \left\{ \frac{0.3}{20} + \frac{0.4}{40} + \frac{0.7}{60} + \frac{0.4}{80} + \frac{0.2}{100} \right\}.$$

For the fuzzy sets operation:

(a) $A_{\sim} \cup B_{\sim}$ (b) $A_{\sim} \cap B_{\sim}$ (c) $\bar{A}_{\sim}$ (d) $A_{\sim} / B_{\sim}$ (e) $\overline{A_{\sim} \cup B_{\sim}}$ (f) $\overline{A_{\sim} \cap B_{\sim}}$

4.1. Two fuzzy sets $\underline{\underline{A}}$ and $\underline{\underline{B}}$, both defined on X , are as follows:

$\mu(x_i)$	x_1	x_2	x_3	x_4	x_5	x_6
$\underline{\underline{A}}$	0.1	0.7	0.8	1.0	0.7	0.1
$\underline{\underline{B}}$	1.0	0.9	0.5	0.2	0.1	0

Express the following λ -cut sets using Zadeh's notation:

- | | |
|--|--|
| (a) $(\overline{\underline{\underline{A}}})_{0.6}$, | (e) $(\underline{\underline{A}} \cup \overline{\underline{\underline{A}}})_{0.7}$, |
| (b) $(\underline{\underline{B}})_{0.5}$, | (f) $(\underline{\underline{B}} \cap \overline{\underline{\underline{B}}})_{0.6}$, |
| (c) $(\underline{\underline{A}} \cup \underline{\underline{B}})_{0.8}$, | (g) $(\overline{\underline{\underline{A}} \cap \underline{\underline{B}}})_{0.8}$, |
| (d) $(\underline{\underline{A}} \cap \underline{\underline{B}})_{0.6}$, | (h) $(\overline{\underline{\underline{A}}} \cup \overline{\underline{\underline{B}}})_{0.7}$. |

4.3. The fuzzy sets $\tilde{A}$, $\tilde{B}$, and $\tilde{C}$ are all defined on the universe $X = [0, 5]$ with the following membership functions:

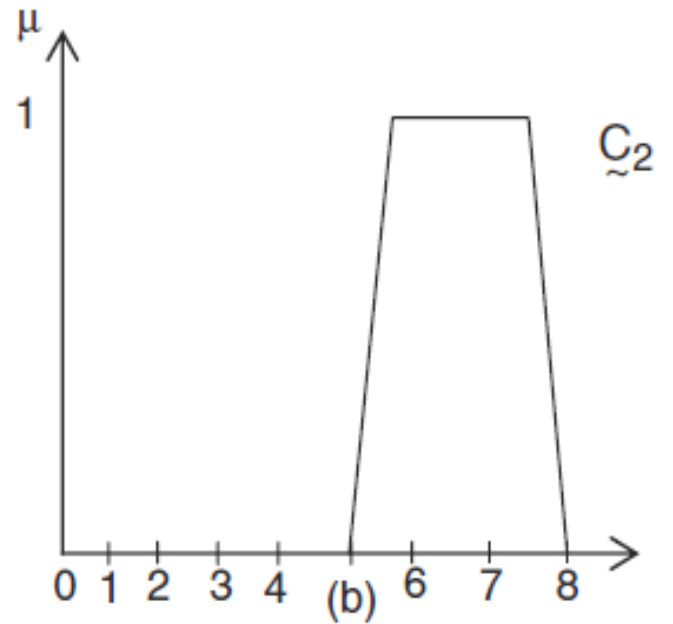
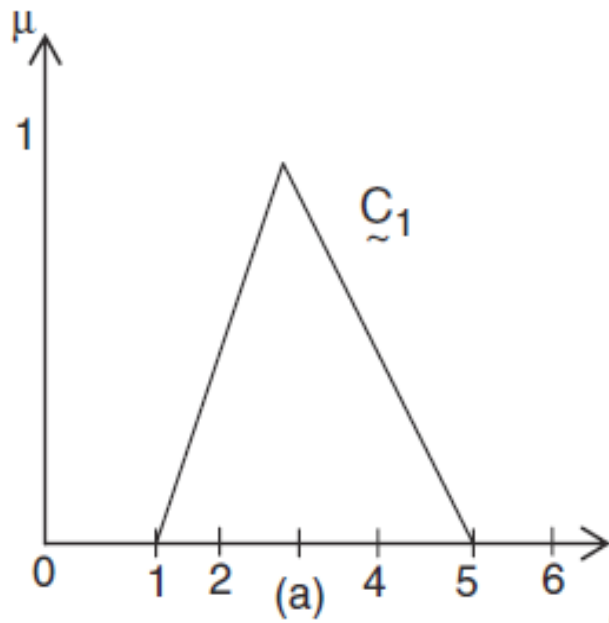
$$\mu_{\tilde{A}}(x) = \frac{1}{1 + 5(x - 5)^2}, \quad \mu_{\tilde{B}}(x) = 2^{-x}, \quad \text{and} \quad \mu_{\tilde{C}}(x) = \frac{2x}{x + 5}.$$

- (a) Sketch the membership functions.
- (b) Define the intervals along the x axis corresponding to the λ -cut sets for each of the fuzzy sets $\tilde{A}$, $\tilde{B}$, and $\tilde{C}$ for the following values of λ :
 - (i) $\lambda = 0.2$
 - (ii) $\lambda = 0.6$
 - (iii) $\lambda = 0.9$
 - (iv) $\lambda = 1.0$.

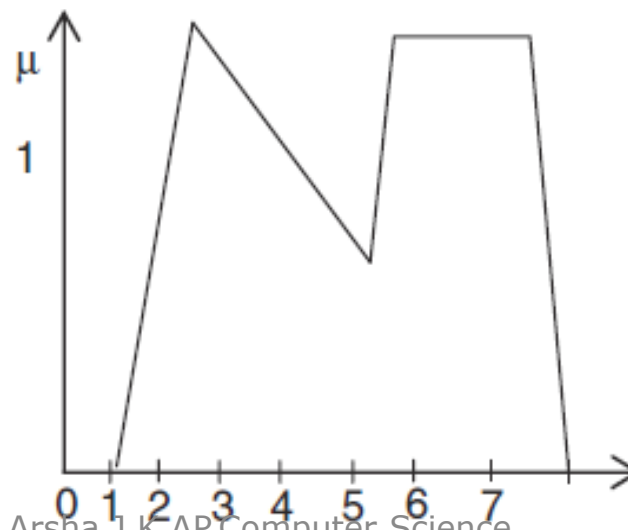
4.4. Determine the crisp λ -cut relations for $\lambda = 0.1j$, for $j = 0, 1, \dots, 10$, for the following fuzzy relation matrix $\tilde{R}$:

$$\tilde{R} = \begin{bmatrix} 0.2 & 0.7 & 0.4 & 1 \\ 1 & 0.9 & 0.5 & 0.1 \\ 0 & 0.8 & 1 & 0.6 \\ 0.2 & 0.5 & 1 & 0.3 \end{bmatrix}.$$

- Apart from the lambda cut sets and relations which convert fuzzy sets or relations into crisp sets or relations, there are other various defuzzification methods employed to convert the fuzzy quantities into crisp quantities.
- The output of an entire fuzzy process can be union of two or more fuzzy membership functions.
- To explain this in detail, consider a fuzzy output, which is formed by two parts, one part being triangular shape and other part being trapezoidal.
- The union of these two forms the outer envelop of the two shapes.



$$C = C_1 \cup C_2$$

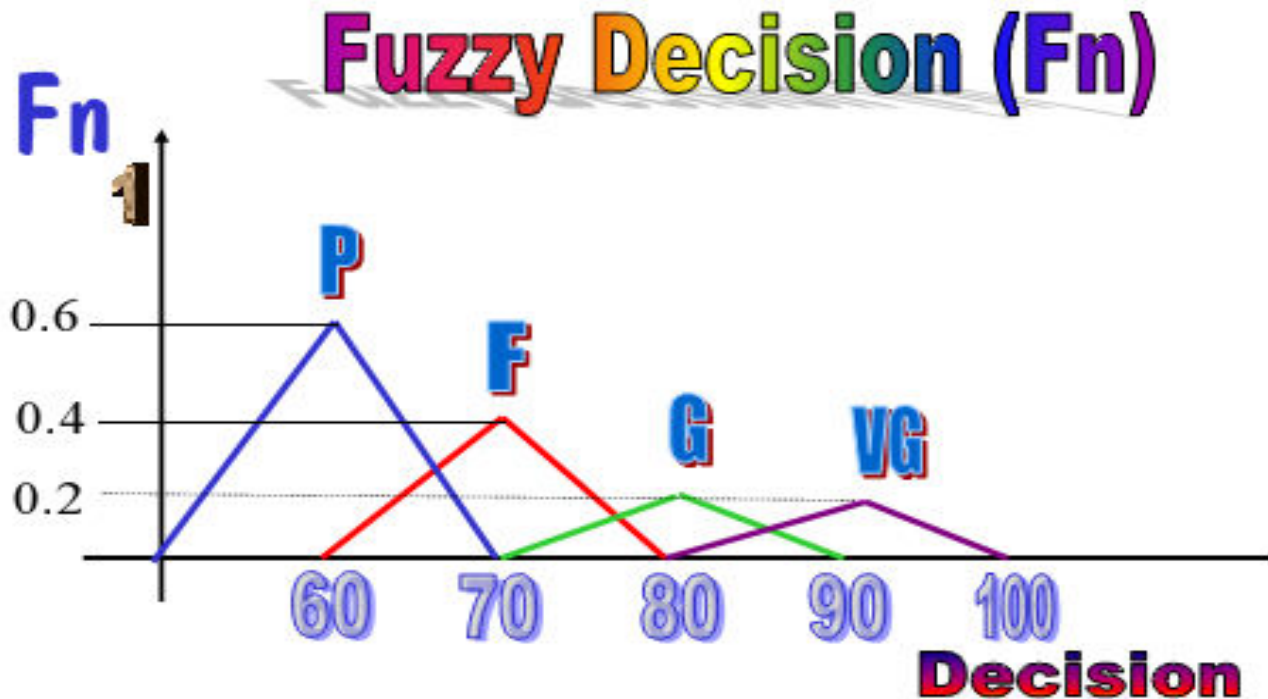


METHODS OF DEFUZZIFICATION

Defuzzification is the process of conversion of a fuzzy quantity into a precise quantity. Defuzzification methods include:

- Max-membership principle,
- Centroid method,
- Weighted average method,
- Mean-max membership,
- Center of sums,
- Center of largest area,
- First of maxima, last of maxima.

FUZZY DECISION



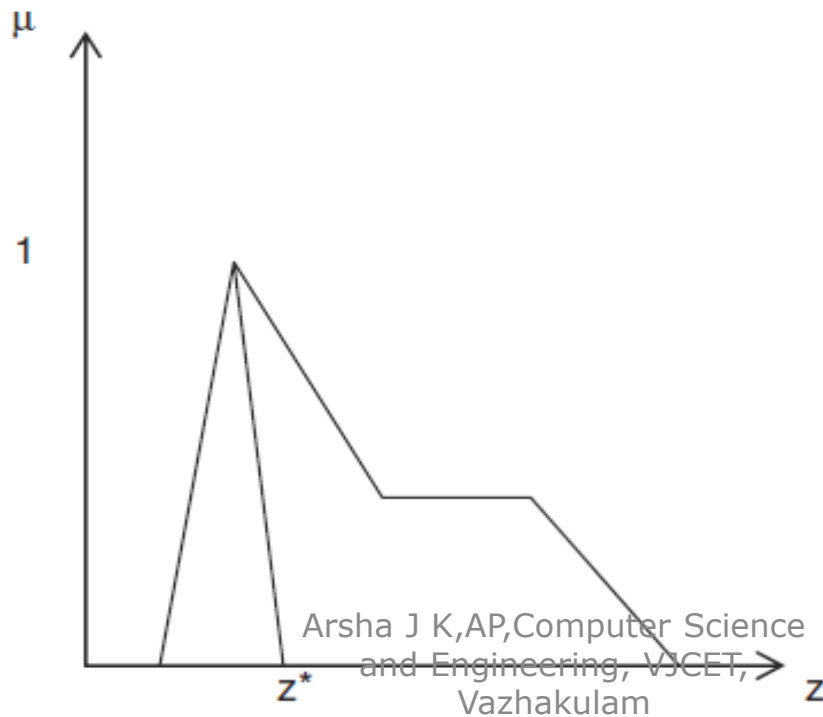
$$F_n = \{P, F, G, VG, E\}$$
$$F_n = \{0.6, 0.4, 0.2, 0.2, 0\}$$

MAX MEMBERSHIP METHOD

This method is given by the expression,

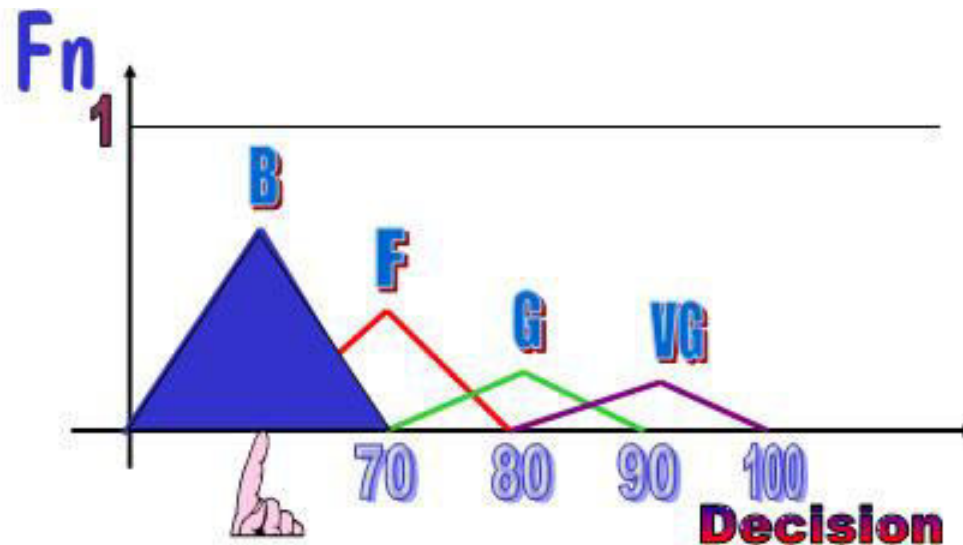
$$\mu_{\tilde{C}}(z^*) \geq \mu_{\tilde{C}}(z) \quad \text{for all } z \in \mathcal{Z}.$$

This method is also referred as height method. This is shown in Fig.



MAX MEMBERSHIP METHOD

- Fuzzy set with the largest membership value is selected.
- Fuzzy decision: $F_n = \{P, F, G, VG, E\}$
- $F_n = \{0.6, 0.4, 0.2, 0.2, 0\}$
- Final decision (FD) = Poor Student
- If two decisions have same membership max, use the average of the two.

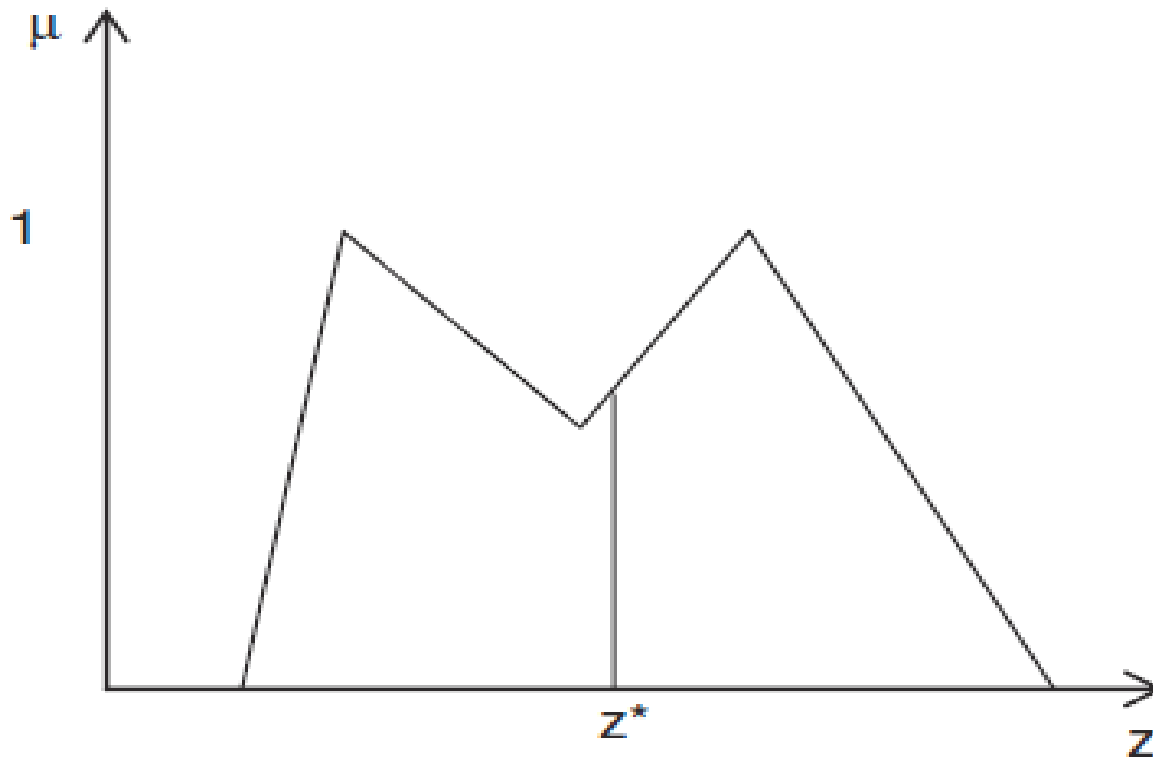


CENTROID METHOD

This method is also known as center-of-mass, center-of-area, or center-of-gravity method. It is the most commonly used defuzzification method. The defuzzified output x^* is defined as

$$x^* = \frac{\int \mu_{\tilde{C}}(x) \cdot x dx}{\int \mu_{\tilde{C}}(x) dx}$$

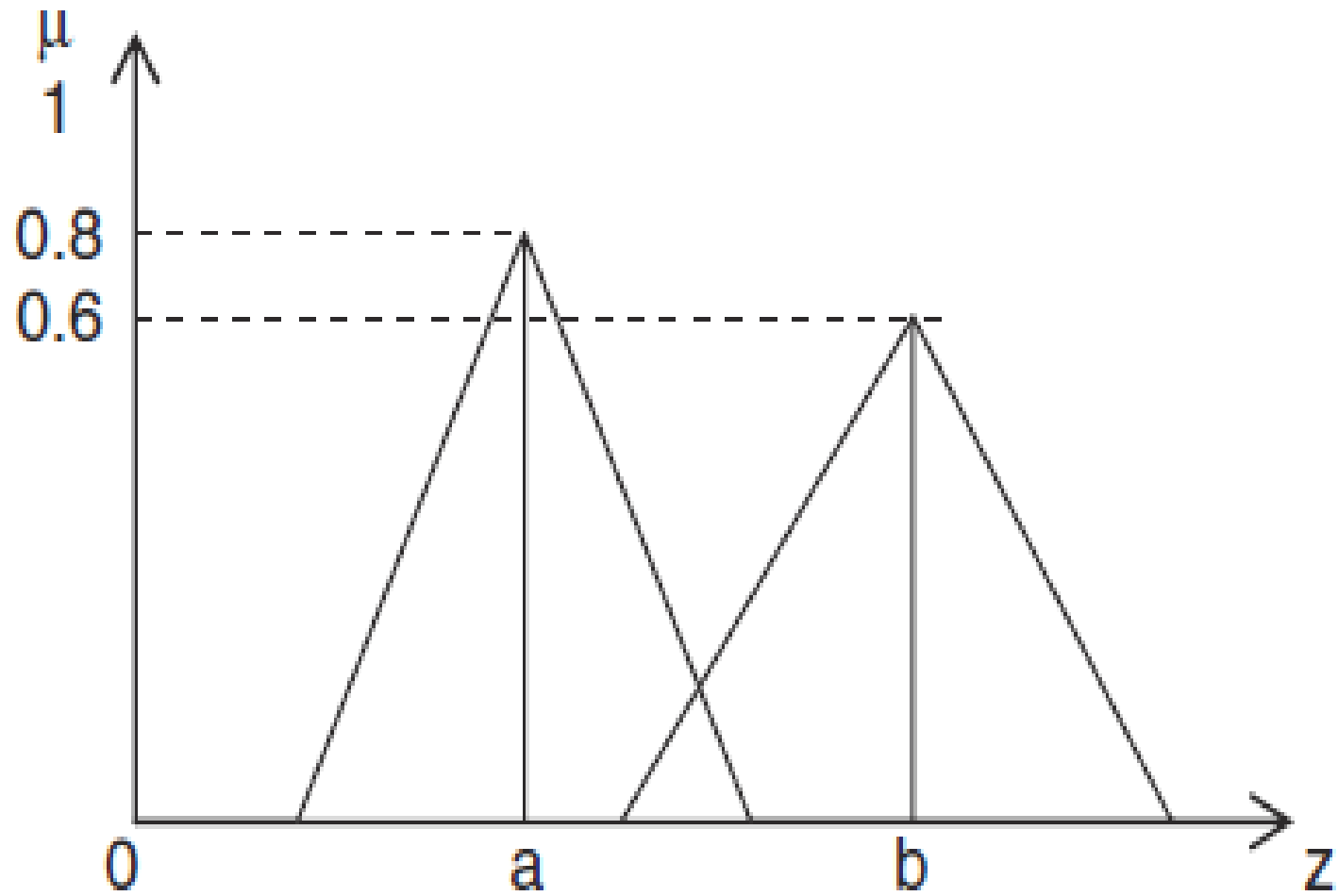
where the symbol $\int$ denotes an algebraic integration.



WEIGHTED AVERAGE METHOD

- This method cannot be used for asymmetric membership functions
- Can only be used for symmetric output membership functions
- Weighing each membership function in the obtained output by its largest membership value forms this method

$$z^* = \frac{\sum_{\sim} \mu_C(\bar{z}) \bar{z}}{\sum_{\sim} \mu_C(\bar{z})},$$

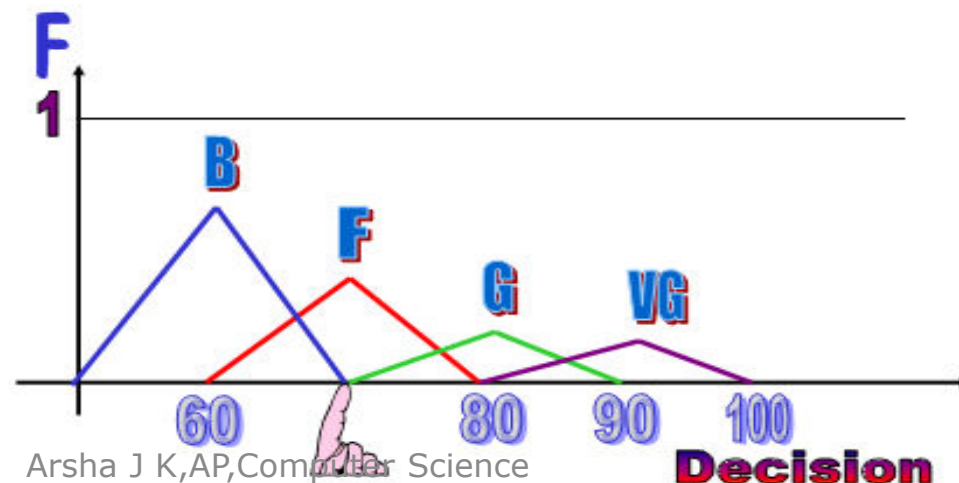


WEIGHTED AVERAGE METHOD

$$FD = \frac{\sum_i \mu_f \cdot fn}{\sum_i \mu_f} = \frac{\mu_E \times E + \mu_{VG} \times VG + \dots}{\mu_E + \mu_{VG} + \dots}$$

$$FD = \frac{0 \times 100 + 0.2 \times 90 + 0.2 \times 80 + 0.4 \times 70 + 0.6 \times 60}{0.2 + 0.2 + 0.4 + 0.6} = 70$$

Final Decision (FD) = Fair Student



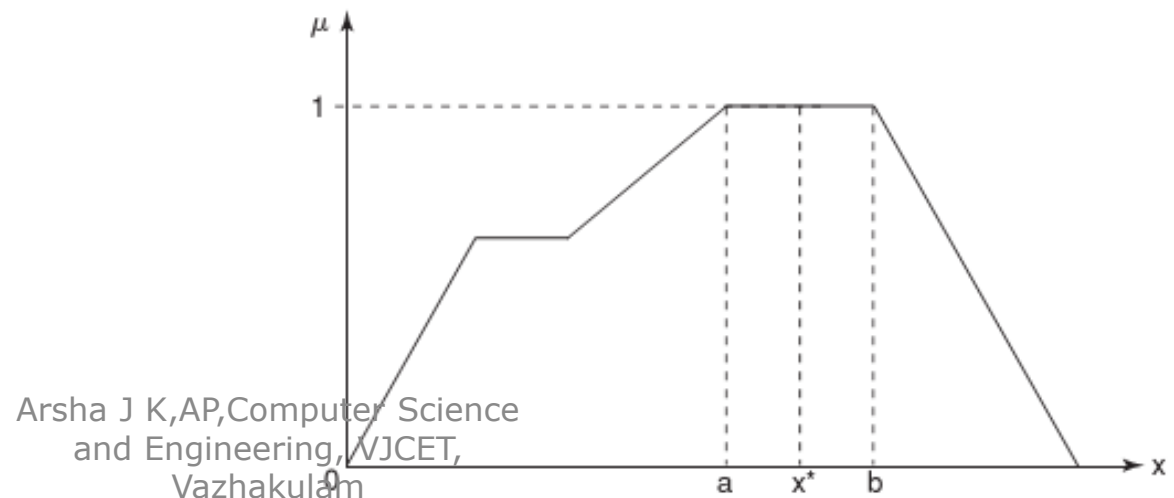
MEAN MAX MEMBERSHIP METHOD

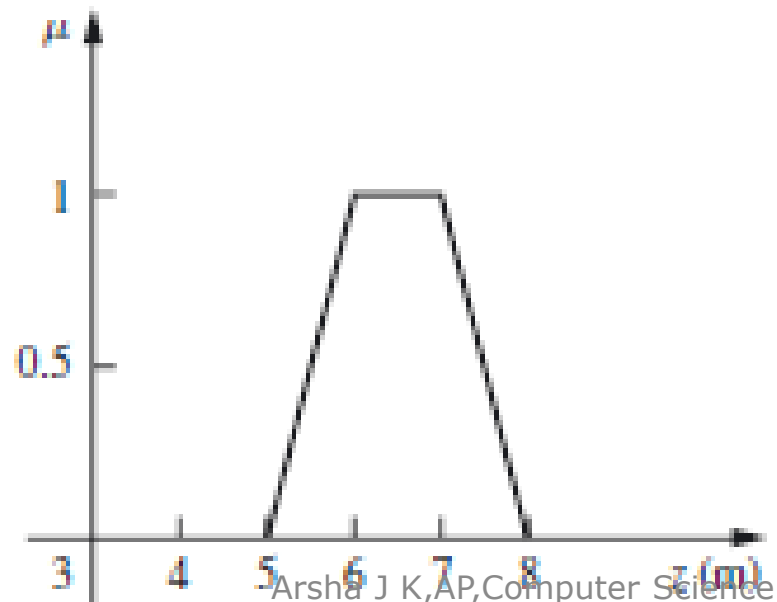
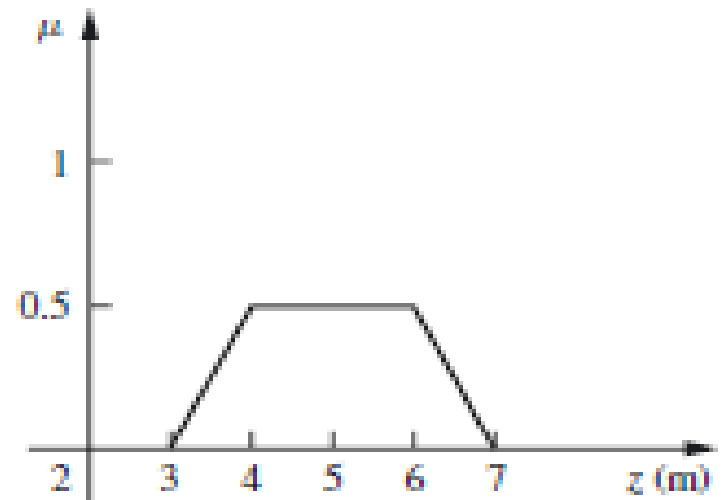
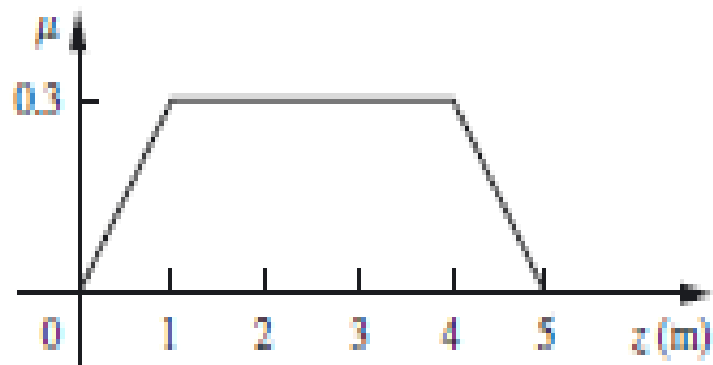
This method is related to max-membership principle, but the present of the maximum membership need not be unique, i.e., the maximum membership need not be a single point, it can be a range. This method is also called as middle of maxima method the expression is given as

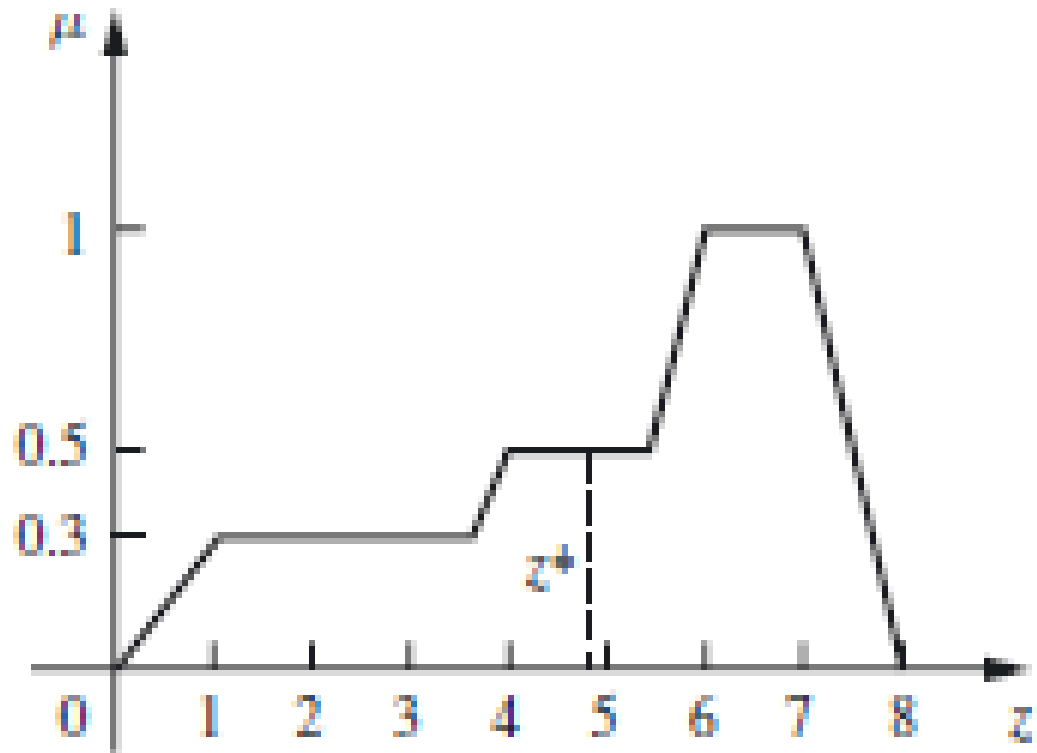
$$z^* = \frac{a + b}{2},$$

where $a \times b$ are the end point of the maximum membership range as shown in Fig. 5.5.

$$x^* = \frac{a + b}{2}$$







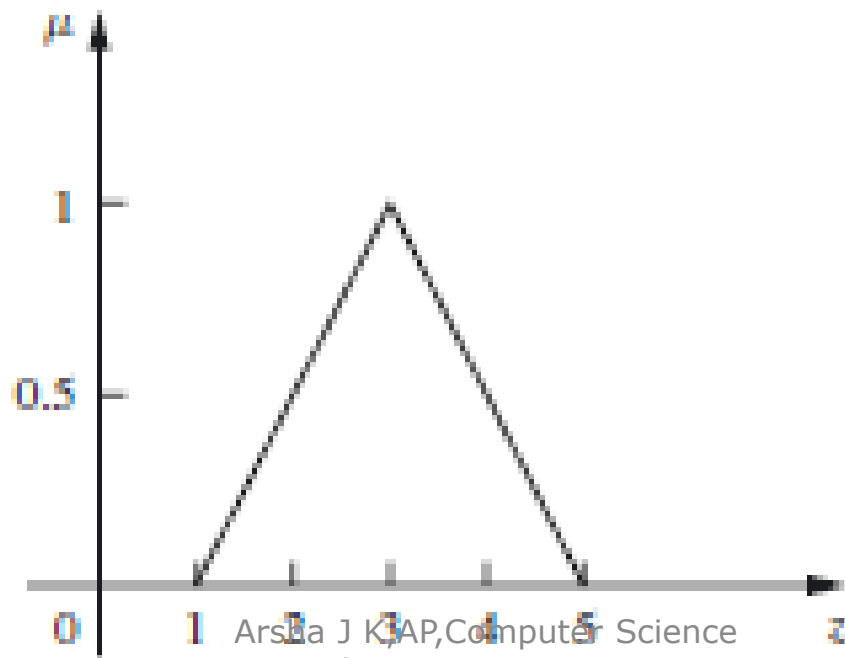
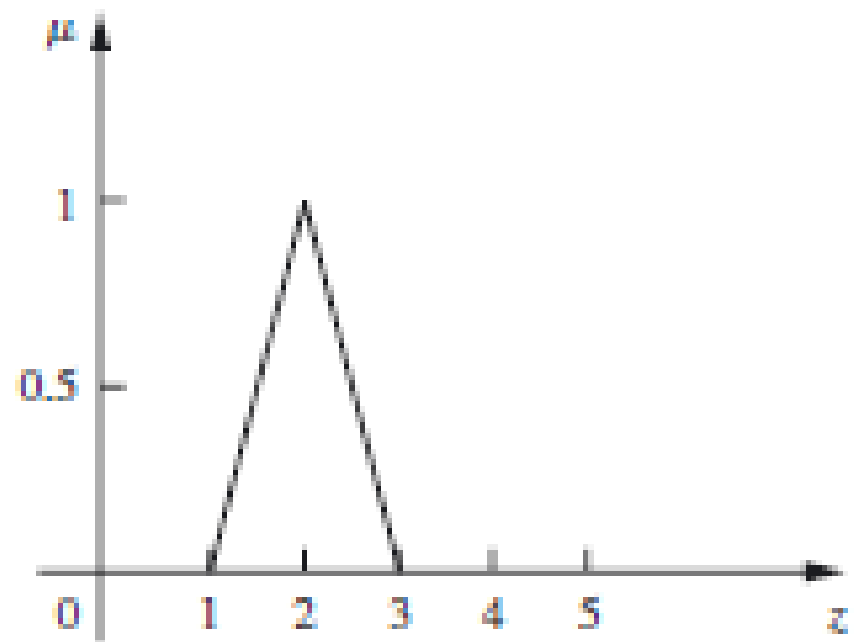
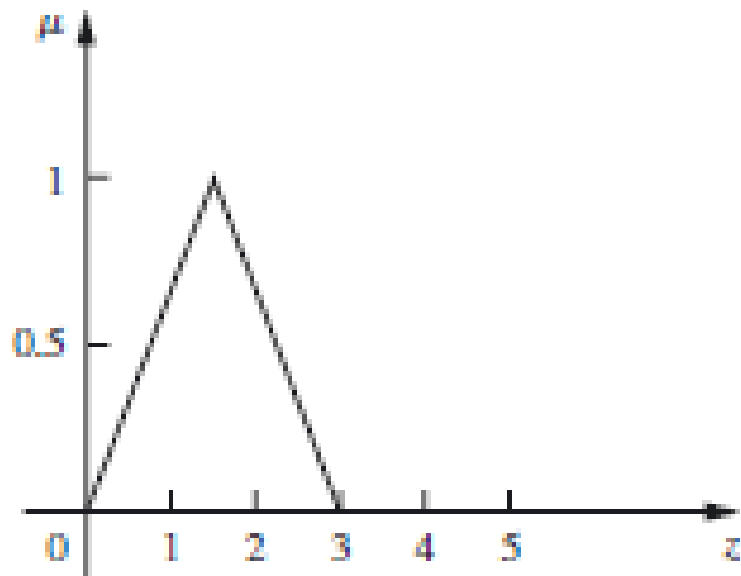
According to the centroid method, Equation (4.5), z^* can be found using

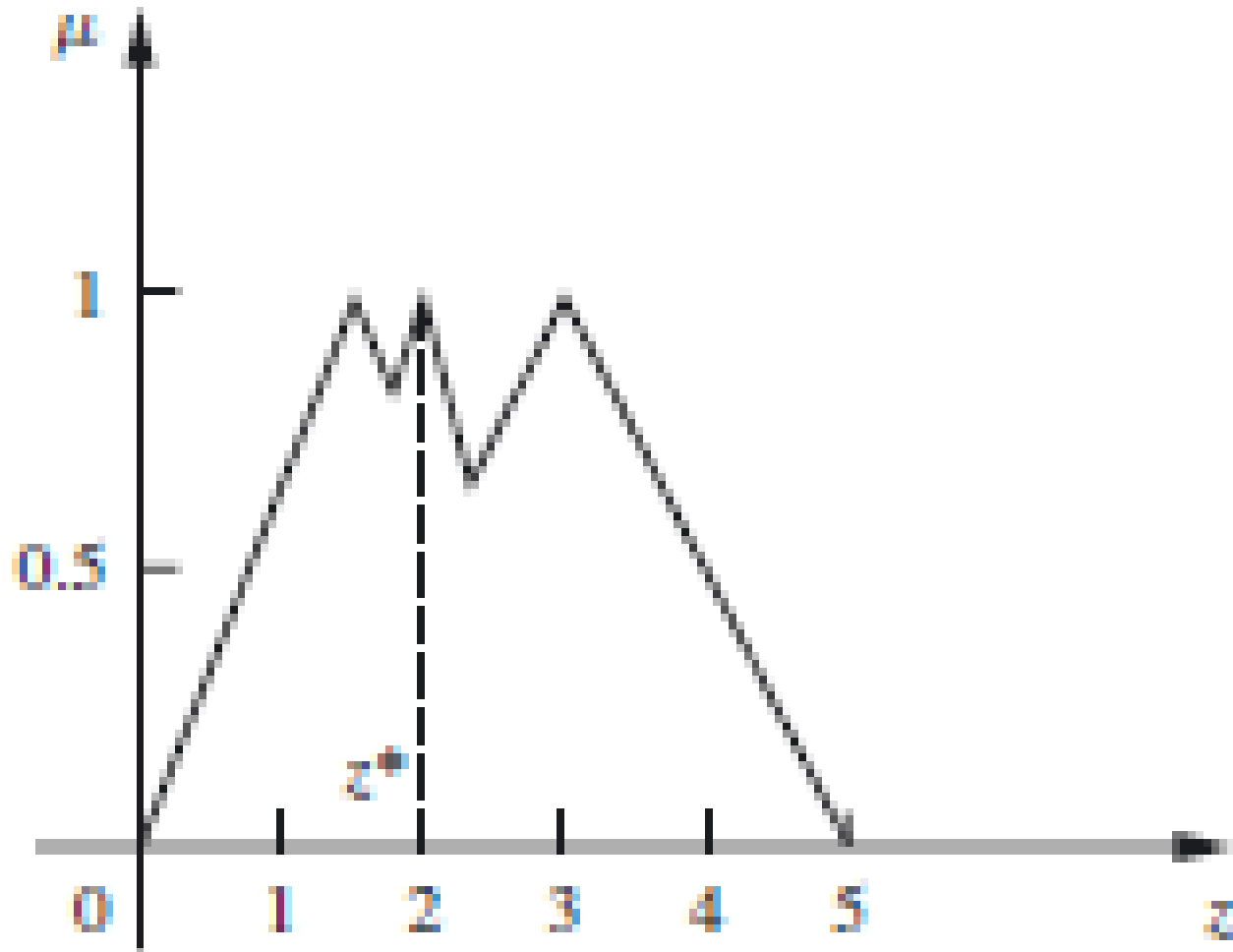
$$\begin{aligned}
 z^* &= \frac{\int \mu_{\underline{u}}(z) \cdot z \, dz}{\int \mu_{\underline{u}}(z) \, dz} \\
 &= \left[\int_0^1 (0.3z)z \, dz + \int_1^{3.6} (0.3)z \, dz + \int_{3.6}^4 \left(\frac{z-3.0}{2} \right) z \, dz + \int_4^{5.5} (0.5)z \, dz \right. \\
 &\quad \left. + \int_{5.5}^6 (z-5)z \, dz + \int_6^7 z \, dz + \int_7^8 (8-z)z \, dz \right] \\
 &\div \left[\int_0^1 (0.3z) \, dz + \int_1^{3.6} (0.3) \, dz + \int_{3.6}^4 \left(\frac{z-3.6}{2} \right) \, dz + \int_4^{5.5} (0.5) \, dz \right. \\
 &\quad \left. + \int_{5.5}^6 \left(\frac{z-5.5}{2} \right) \, dz + \int_6^7 \, dz + \int_7^8 \left(\frac{7-z}{2} \right) \, dz \right] \\
 &= 4.9 \text{ m,}
 \end{aligned}$$

where z^* is shown in Figure 4.19. According to the weighted average method, Equation (4.6),

$$z^* = \frac{(0.3 \times 2.5) + (0.5 \times 5) + (1 \times 6.5)}{0.3 + 0.5 + 1} = 5.41 \text{ m,}$$

and is shown in Figure 4.20. According to the mean max membership method, Equation (4.7), z^* is given by $(6 + 7)/2 = 6.5 \text{ m}$, and is shown in Figure 4.21.





According to the centroid method, Equation (4.5),

$$\begin{aligned} z^* &= \frac{\int \mu_B(z) z \, dz}{\int \mu_B(z) \, dz} \\ &= \left[\int_0^{1.5} (0.67z) z \, dz + \int_{1.5}^{1.8} (2 - 0.67z) z \, dz + \int_{1.8}^2 (z - 1) z \, dz + \int_2^{2.33} (3 - z) z \, dz \right. \\ &\quad \left. + \int_{2.33}^3 (0.5z - 0.5) z \, dz + \int_3^5 (2.5 - 0.5z) z \, dz \right] \\ &\div \left[\int_0^{1.5} (0.67z) \, dz + \int_{1.5}^{1.8} (2 - 0.67z) \, dz + \int_{1.8}^2 (z - 1) \, dz + \int_2^{2.33} (3 - z) \, dz \right. \\ &\quad \left. + \int_{2.33}^3 (0.5z - 0.5) \, dz + \int_3^5 (2.5 - 0.5z) \, dz \right] \\ &= 2.5 \text{ m.} \end{aligned}$$

According to the weighted average method, Equation (4.6),

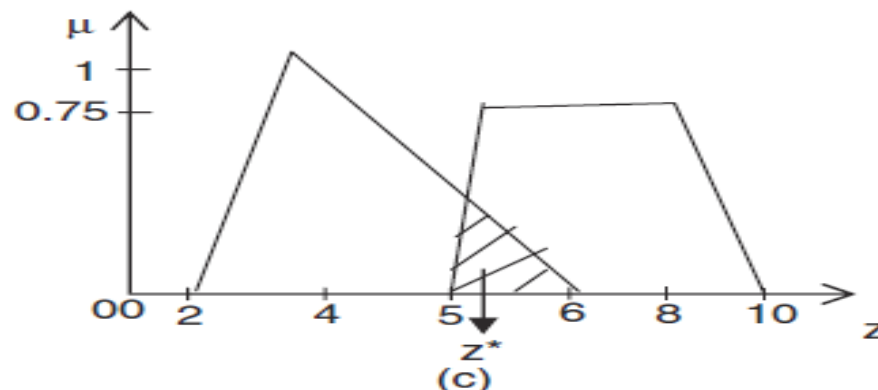
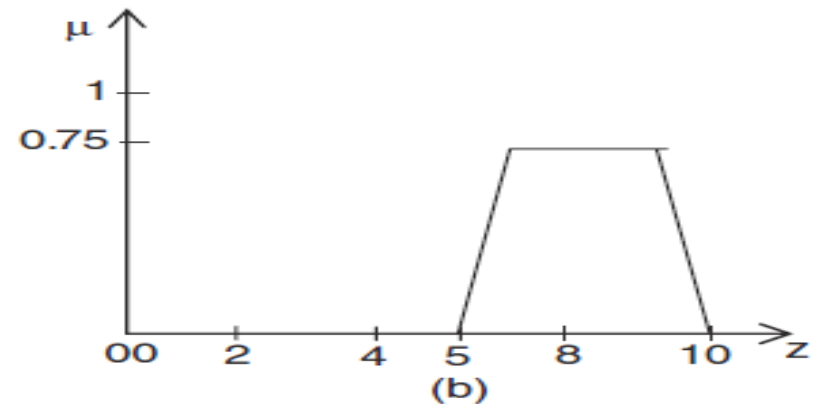
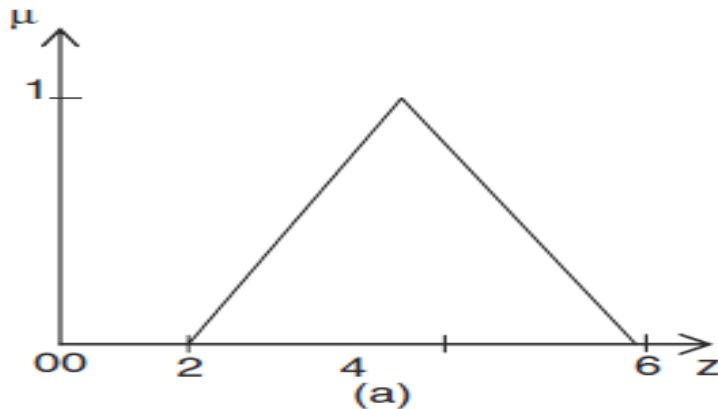
$$z^* = \frac{(1 \times 1.5) + (1 \times 2) + (1 \times 3)}{1 + 1 + 1} = 2.25 \text{ m,}$$

and is shown in Figure 4.27.

CENTER OF SUMS

This method employs the algebraic sum of the individual fuzzy subsets instead of their unions. The calculations here are very fast but the main drawback is that the intersecting areas are added twice. The defuzzified value x^* is given by

$$x^* = \frac{\int_x x \sum_{i=1}^n \mu_{G_i}(x) dx}{\int_x \sum_{i=1}^n \mu_{G_i}(x) dx}$$

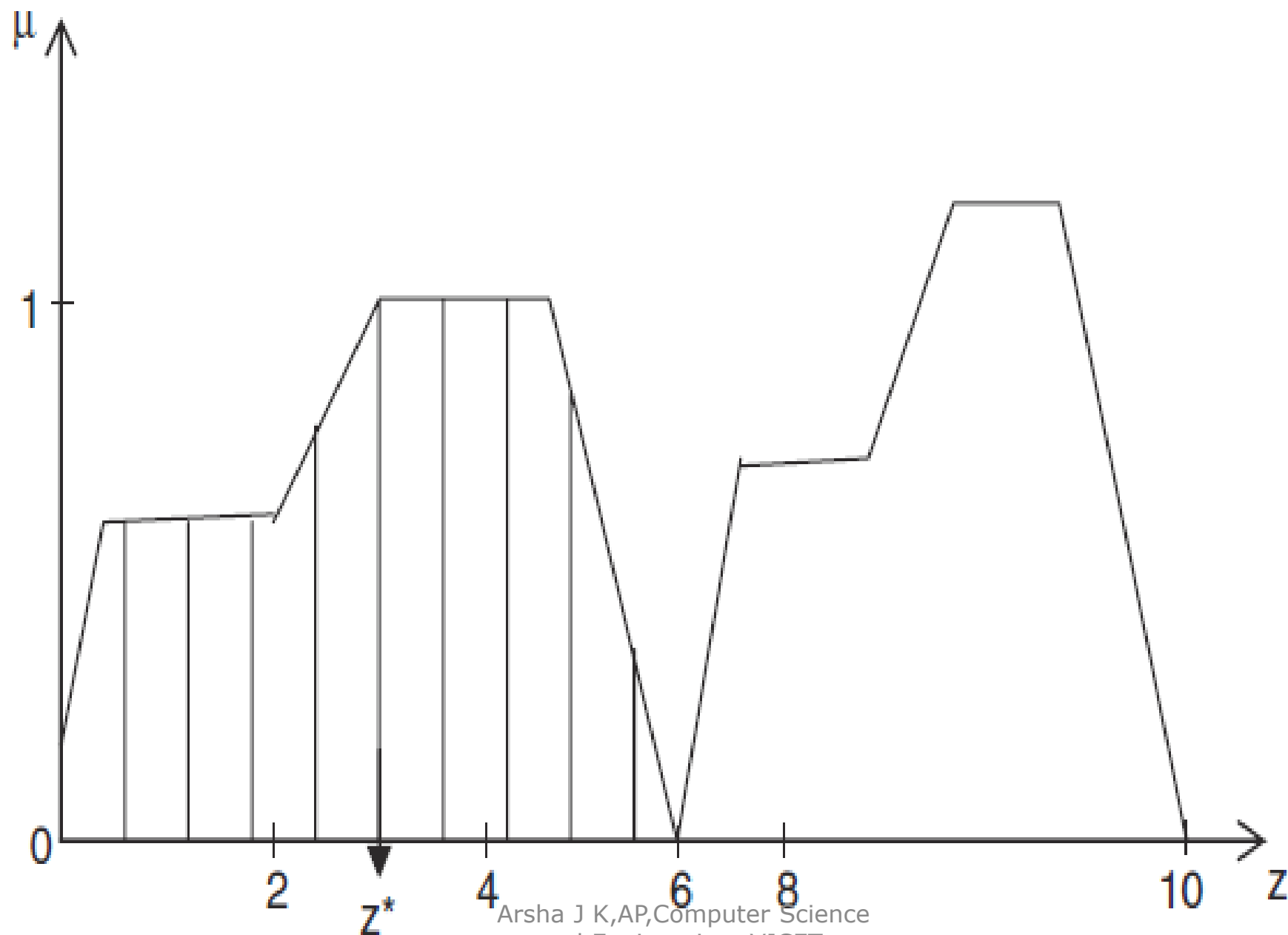


CENTER OF LARGEST AREA

This method can be adopted when the output consists of at least two convex fuzzy subsets which are not overlapping. The output in this case is biased towards a side of one membership function. When output fuzzy set has at least two convex regions then the center-of-gravity of the convex fuzzy subregion having the largest area is used to obtain the defuzzified value x^* . This value is given by

$$x^* = \frac{\int \mu_{c_i}(x) \cdot x dx}{\int \mu_{c_i}(x) dx}$$

where c_i is the convex subregion that has the largest area making up C_j



FIRST OF MAXIMA (LAST OF MAXIMA)

The steps used for obtaining crisp values are as follows:

1. Initially, the maximum height in the union is found:

$$\text{hgt}(\tilde{c}_j) = \sup_{x \in X} \mu_{\tilde{c}_j}(x)$$

where sup is supremum, i.e., the least upper bound.

2. Then the first of maxima is found:

$$x^* = \inf_{x \in X} \{x \in X \mid \mu_{\tilde{c}_j}(x) = \text{hgt}(\tilde{c}_j)\}$$

where inf is the infimum, i.e., the greatest lower bound.

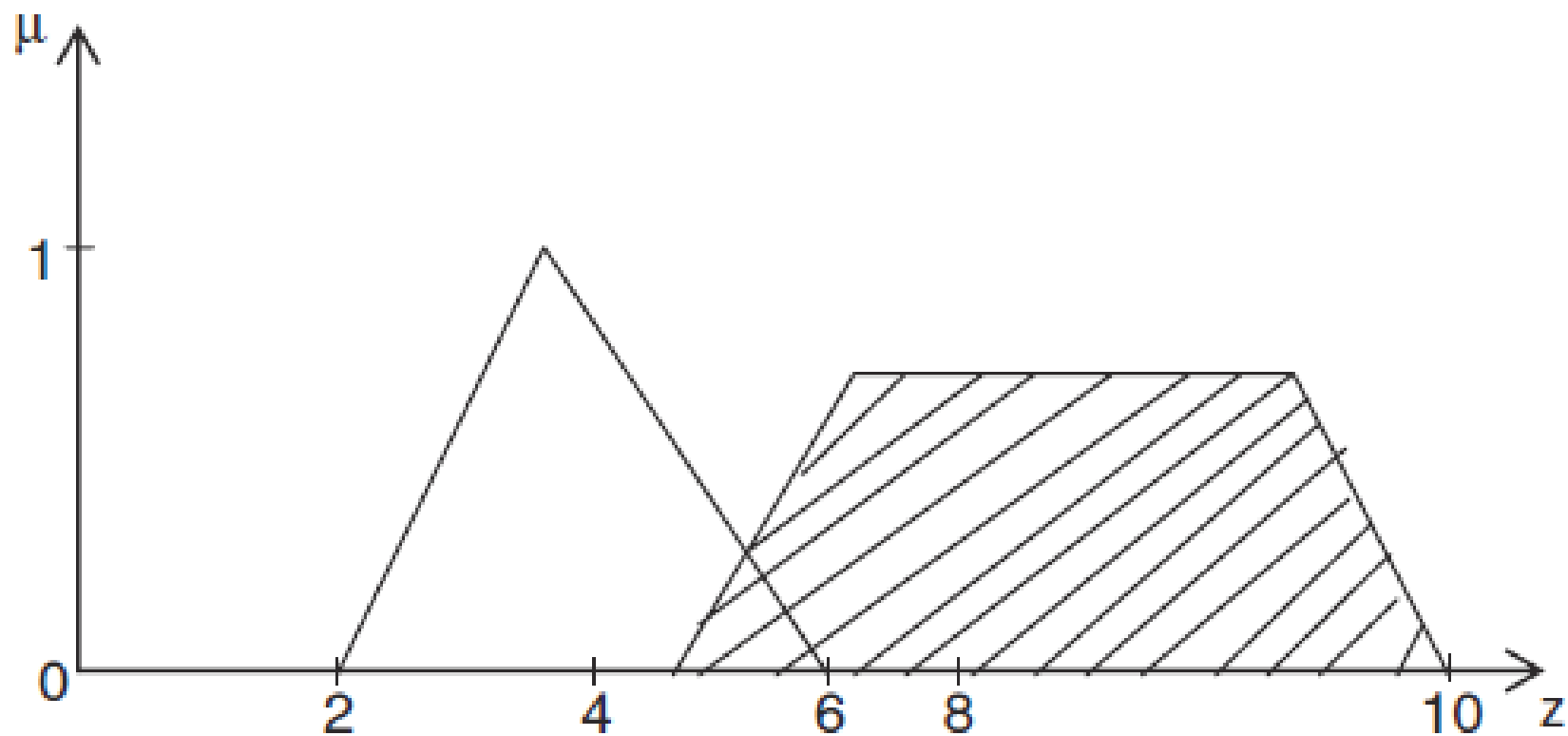
3. After this the last maxima is found:

$$x^* = \sup_{x \in X} \{x \in X \mid \mu_{\tilde{c}_j}(x) = \text{hgt}(\tilde{c}_j)\}$$

where

sup = supremum, i.e., the least upper bound

inf = infimum, i.e., the greatest lower bound



SUMMARY

- Defuzzification process is essential because some engineering applications need exact values for performing the operation.
- Example: If speed of a motor has to be varied, we cannot instruct to raise it “slightly”, “high”, etc., using linguistic variables; rather, it should be specified as raise it by 200 rpm or so, i.e., a specific amount of raise should be mentioned.
- The method of defuzzification should be assessed on the basis of the output in the context of data available.